A330

VER. 26.10.24

KneeBoard

Click for Update

By Flyingdeuk

DOMESTIC

JAPAN

CHINA

SE ASIA

OCEANIA

MONGO, ME

Differences

FUEL

1 ENG TAXI

LP/HP Air

HP Air Start

Cold Temp

Cold Wx

ENG ON Deicing

ENG OFF Deicing

QRH

RESET

복행 GA 이후 / Go Around In flight

손님 여러분, 저는 기장입니다.
우리 비행기는 ◐_____ 다시 상승했습니다.

◐ 기류가 불안정해
◐ 관제탑의 지시로
◐ 활주로가 육안으로 확인되지 않아
◐ 기술적인 이유로

[재접근 가능 시]

◐_____, 잠시 후에 착륙할 예정입니다. 기다려 주셔서 고맙습니다.

◐ 기상 상태가 좋아져서
◐ 관제탑으로부터 다시 착륙을 허가 받아
◐ 관련된 문제가 해결되어

[Holding/Diversion 필요시]

이 점 양해해 주시기 바랍니다.

Ladies and gentlemen, this is the captain speaking.
We performed a go-around ◐ _____.

◐ due to unstable winds
◐ because we could not visually confirm the runway
◐ due to instructions from Air Traffic Control
◐ for technical reasons

[Re-approach]

◐ _____ , and we expect to land shortly. Thank you for your understanding.

◐ The weather has improved
◐ We have now received clearance
◐ The situation has been resolved

[Holding/Diversion is required]

Thank you for your patience and understanding.



DOMESTIC

ICN 인천국제

GMP 김포국제

CJU 제주국제

PUS 김해국제

WELCOME PA (10분이상 지연 or 예상시 지연 방송)

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___ (국제)공항까지 비행시간은 ___시간 ___분 으로 예상됩니다.
비행 중에는 항공기가 갑자기 흔들릴 수도 있으니, 자리에 앉아 계실 때에는 항상 좌석벨트를 매주시기 바랍니다. (저희 승무원 모두는)
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Good morning (afternoon /evening), ladies and gentlemen.
This is captain last name speaking.
Welcome aboard Korean Air.
This flight is bound for ___(international) airport and our flight time is ___ hours(s) and minutes.
For your safety, keep your seatbelts fastened while you are seated.
Thank you for choosing Koreanair.
Please enjoy your flight.



도착방송

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

동남아 지역 난기류 대비 기내 서비스 절차

- **6**시간 미만 중거리만 **1**시간 전 음료서비스
- **40**분전 서비스 회수
- **APP PA**이후 **CREW** 착석 강조

손님 여러분, 저는 기장입니다.
우리 비행기는 앞으로 약 (40)분 후에 ___국제공항에 착륙 예정입니다.
현재 공항의 날씨는 ◐___, 기온은 섭씨 __도 입니다.

◐ 맑으며
◐ (다소)흐리며
◐ (이슬)비가 내리며/소나기가 내리며
◐ 바람이 불고 있으며
◐ 눈이 오고 있으며
◐ 안개가 끼어 있으며
◐ 황사가 있으며

지금 이곳의 시각은 __월 __일 __요일, 오전(오후) __시 __분 입니다. 고맙습니다.

Ladies and gentlemen, this is the captain speaking.
We expect to land at __ international airport in about (40) minutes.
The current temperature at ___ is __ degrees Celsius, or __ degrees Fahrenheit (NAVBLUE 참조)
and it is ◐___.

◐ (mostly) clear
◐ (partly) cloudy
◐ drizzling / raining
◐ windy
◐ snowing
◐ foggy
◐ hazy or smoggy

The current time is __ : __ a.m(p.m), on (day-of-the-week), (month)(date).



JAPAN

HND	KIX	NRT
도쿄 하네다	오사카 간사이	도쿄 나리타
FUK	**NGO**	**CTS**
후쿠오카	나고야 주부 (Chubu Centrair)	삿포로 신(New) 치토세
KOJ		
가고시마		

WELCOME PA (10분이상 지연 or 예상시 지연 방송)

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CHINA

HKG	**PEK**	**WUH**
홍콩	베이징 소우뚜	우한 텐허
CAN	**XIY**	**TSN**
광저우 바이윈	시안 시엔양	텐진 빈하이
TAO	**SZX**	**PVG**
칭다오 자오둥	선전 바오안	샹하이 푸동
SHE		
선양 타오쎈		

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SE ASIA

BKK	**SIN**	**DAD**
방콕 수완나폼	싱가포르 창이	다낭
SGN	**CNX**	**DPS**
호찌민 탄소넛	치앙마이	덴파사르 발리 응우라라이
KTI	**TPE**	**RGN**
프놈펜 테초	타이페이 타오유엔	양곤
KUL	**CXR**	**PQC**
쿠알라룸푸트	나짱(나트랑)/깜라인(깜란)	푸꾸옥
HAN	**MNL**	
하노이 노이바이	마닐라 니노이아키노	

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도착방송

OCEANIA

SYD BNE

시드니
킹스폴드 스미스

브리즈번

WELCOME PA (10분이상 지연 or 예상시 지연 방송)

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도착방송

MONGO, ME ASIA

UBN 울란바타르 칭기스칸

DXB 두바이

KTM 카트만두 트리부반

WELCOME PA (10분이상 지연 or 예상시 지연 방송)

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도착방송

RKSI(ICN) 23ft

KE ICN 131.5
(TOBT 조절)

UTC+9

DCL -10min
TOBT 5분차이시 CTC COMM

ARR
TL140

Tx PT GUKDO, OLMEN, REBIT, KARBU
FL180,170,160 예상

33L/R(334)	OLMEN	xE	ENPIL	ILS
34L/R(334)	GUKDO	xY	PAMBI	ILS
15L/R(154)	REBIT	xC	BITIM	ILS
16L/R(154)	KARBU	xH	MUNAN	ILS Z

NCN 113.8 33L INLL 109.3 33R INRR 108.9 15L ISLL 111.9 15R ISRR 109.1
WNG 112.9 34L IRFN 109.95 34R IRKN 108.1 16L IRKS 110.35 16R IRFS 108.35

SPD 8NM 180kts, 5NM 160kts

FIX YJU R271, P73 /2, P518 R068, R278

33R : C5(5501'), C6(6683'), 33L : B4(7463'), B5(8513')
15L : C4(5495'), C5(6227'), 15R : B3(7454'), B2(8641') HIRO

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') HIRO

Parallel TWY 10KTS이상, R17 Max 15KTS

DEP
TA140

NADP
RWY 33/34 NADP 1, RWY 15/16 NADP 2

33L/R(334)	BOPTA	xE/A	ALT ATC(5500)
34L/R(334)	EGOBA	xY	ALT ATC
15L/R(154)	NOPIK (15/16 : BINIL)	xC	ALT 5000
16L/R(154)	OSPOT	xH	ALT 5000

EO 33L/R : NC05L/R R242,
34L/R : EO34L/R R242

Parallel TWY 10KTS이상, R17 Max 15KTS



PA

RKSS(GMP) 59ft

KE GMP 131.15 UTC+9

DCL -15min
TOBT 5분차이시 CTC COMM

ARR
TL140

RWY 32L Landing
(06:00-09:00L/12:00-15:00L/18:00-21:00)

32L/R (324)	OLMEN GUKDO	xT (xQ)	BUMSI	ILS
14R(144)	KARBU REBIT	xU (xZ)	DOKDO	ILS

NADP
32 : ILS 7 DME L/G D/N, FAF L/D FLAP
14 : LOC CAPT L/G D/N, FAF L/D FLAP

NAV
KIP 113.6 32L IKMO 108.3 32R ISKP 110.7 14L ISEL 109.9 14R IOFR 108.7

FIX
YJU R271, P73 /2, P518 R068, R278

32L : D3(6532'), E2(9117'), 32R(No CL L/T) : E1(6614')
14R : C1(6578')

TWR -> GND -> APRON all by ATC
RTE 1 : D3-P6-N1 H.S P5 (BOKUM, NUNPI, SAMKU)

DEP
TA140

RWY 32R Takeoff
(06:00-09:00L/12:00-15:00L/18:00-21:00)

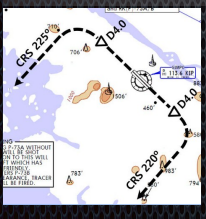
Walk Around
비밀번호 없어짐, 정비사가 열어줌

NADP
FX-10도, CLB, CONF 1+F, ACC 4000FT

32L/R (324)	BULTI EGOBA	xT (xQ)	ALT 5000
14R(144)	NOPIK OSPOT	xU (xZ)	ALT 6000

EO
32L/R : EO32L/R, R225
14L/R : EO14L/R, R220

TWR -> GND -> APRON all by ATC
(BOKUM, NUNPI, SAMKU)



PA

RKPC(CJU) 119ft

KE CJU 129.4 UTC+9 DCL -10min

PREFLT 외부점검 비밀번호 정비사 문의

ARR **AFT Merge PT 220kts, DCT IAF 195kts, FAF 160kts**
TL140

07(066)	DOTOL xP	YUMIN	ILS Z
25(246)	DOTOL xM	DUKAL	ILS Z

RNP Y 07/25 (No LPV), RNP Z 07/25 AR (RNP 0.11)

NAV **YDM 109.0 07 ICJU 109.9 25 ICHE 111.3**

FIX YJU R271, P73 /2, P518 R068, R278

07 : P6(5176'), P5(5882')-HIRO, P4(6840'-Req ATC)
25 : P7(5219'), P8(5882)-HIRO, P10(7524'-Req ATC)

HIRO in Force CHK, HST 40KTS
Entering Rapid TWY CTC GND 121.675 (STOP X)

DEP **07 : Passing G4 CTC TWR**
TA140 **25 : 31 Holding PSN on P,E1,2,3 CTC TWR**

Walk Around 비밀 번호

07(066)	KAMIT xE	ALT 10000
25(246)	KAMIT xW	ALT 10000

EO 25 : YDM 246/3, R290

Caution RW31 Holding Sign & STOP BAR L/T
HIRO in Force Prepare Intersection DEP



PA

RKPK(PUS) 13ft

KE CJU 129.4 UTC+9 DCL -5min

ARR Tail Wind 10kts, 36R 360000lbs 제한
TL140 Circling Procedure Next Page

36L/R (002)	KEVOX x	MASTA	ILS
	9DME L/G D/N 8DME L/D FLAP		
18L/R (182)	GAYHA x	MSATA	VOR Circling

NAV KMH 113.8 36L IKMA 108.5 36R IKHE 109.5

36L : C4 (6299'), C2(7795') 36R : E3(5866'), E2(7339')
18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3, C4 by ATC Only. Max Taxi SPD 20kts
C2 HOLD SHORT 가까움 (VACATE TAXI SPD)

DEP Modified NADP
TA140 CLB 1500, 360000lbs TOGA, CONF 1+F

Walk Around 비밀번호

36L/R (002)	SOORO x KALOD Tx	ALT ATC
18L/R (182)	GIMHAE x	ALT 5000
EO	36 : KMH R091, R271, R185 R290	

RWY36 400ft Manual L/H TURN
Max Taxi SPD 20kts



Circling Next Page

PA

A330 PUS VOR C 18L/R





Input PERF -> SEC COPY

CLR CP18L/R, DISCONT
MDA :[ ], L/D CONF chk

FIX : KMH 280(Base Turn), 284(Missed App)



Missed App

Before MAP : L/H Turn **KMH 284** OUTBD to GAYHA **GA 6000ft**

After MAP : **Before Abeam RWY** HD310 6000ft -> RDR Vector
After Abeam RWY R/H Turn RWY -> HD180, 6000ft -> RDR Vector

A330 PUS LOC 36L/R Circling 18L/R

F-PLN			SEC F-PLN	
APPR	VIA	STAR	APPR	STAR
LOC36L	KEVOX	KEVOX x TRANS MASTA	18L/R	V18L/R
			(MDA: Blank)	
			(LDG CONF chk)	

CI36L(CF36R) 3500 FI36L(FF36R) 2100

FIX : KMH 280(Base Turn), 310(Missed App)



Missed App

Before MAP : L/H Turn **KMH 284** OUTBD to GAYHA **GA 6000ft**

After MAP : **Before Abeam RWY** HD310 6000ft -> RDR Vector
After Abeam RWY R/H Turn RWY -> HD180, 6000ft -> RDR Vector

RJTT(HND) 21ft Terminal 3

None | UTC+9 | DCL -15min

PREFLT 북풍시 34L 주로 사용, 남풍시 22/23사용

ARR TL140

SPENS 220
STAR XAC Night-APP ILS xx Y 1400~
ILS xxZ : 180kts, 160kts limit APP Chart
22/23 LDA, VOR 16L/R Tailored Procedure

34L/R (338)	XAC xK/H XAC xA	KAIHO/CACAO ARLON	ILS X, VIS ILS Z
22(223)	XAC 1B	BACON	LDA W (RNV22-W)
16L/R (158)	XAC R	NATTY/SANDY	RNP (RNV16R/L)
23(231)	XAC 2B	DANON	LDA W (RNV23-W)

NAV

TTE117.4 34L IHA 111.7 16R ITA 111.55 34R
ITC 108.9 16L IOC 111.95
22 IAD 108.1 23ITD 110.5 (LDA 22 IK, 23 ITL)

34L:L12(6515),L13(7165), 16R(DIS):L5(5147),L3(6361)
22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)
34R(DIS):C10(5780),C11(7270), 16L(DIS):C6(5925),C4(7004)

Caution L12,13 then TWY, Gate 105P Procedure

PA

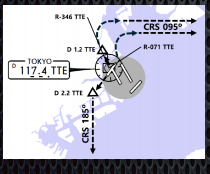
DEP TA140

xx B/C 2200-0230Z 0600-1000Z
34R BEKLA KAIJI 230, TO FLAP SPD Until TORAM
05 OPPAR : Reduce SPD ARC TURN

ALL	BEKLA x OPPAR x	ALT ATC

EO 34L : TTE 346/1.2, R095, 34R : TTE R071, R095, 22 : TTE 223/2.2, R185

RWY05 RTE 5 TAXI Chart



- DEP Freq ATIS

PA

LDA
Next Page

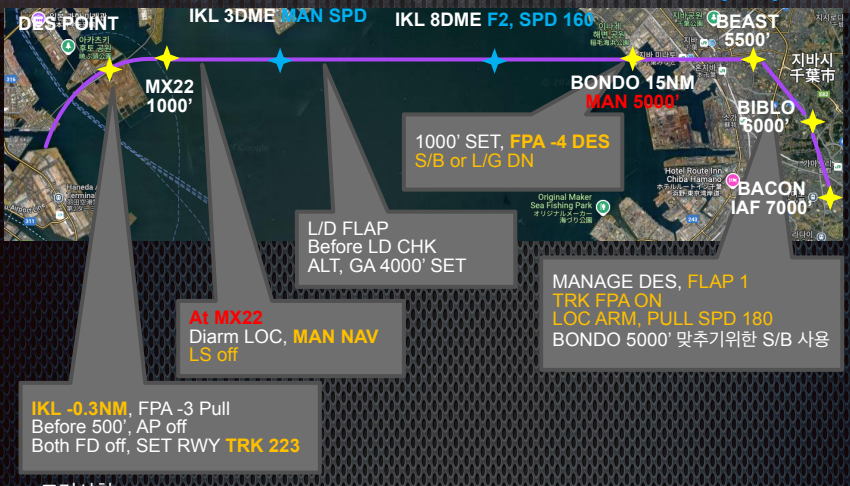


A330 HND LDA 22 W

F-PLN

APPR	STAR	TRANS	FIX
RNV22-W	XAC 1B	BACON	TTE R015

CHK IKL, IKL /8 180kts, /3 160kts, TTE 015 (GA)



고려사항
- Procedure SPD 지켜야 Traffic 간격 유지됨. 무게에 따라 -4도 증속가능성 있음.
- S/B 사용하거나 강하각을 조절하는 것이 필요함.

Missed App

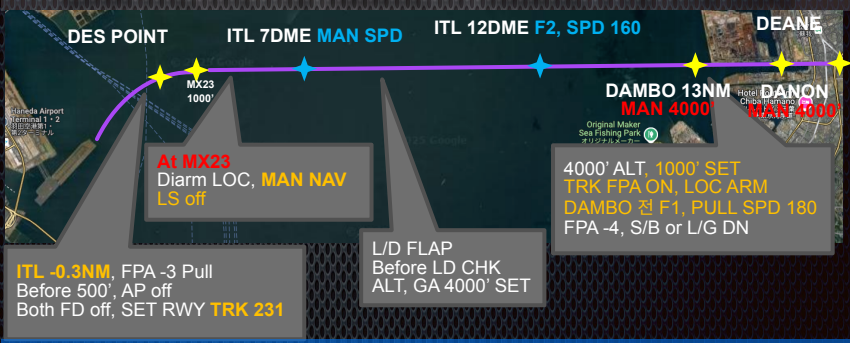
R/H Turn **HME 015** OUTBD (PULL TRK)
(DCT KASGA Radial In 196, MAN NAV)

A330 HND LDA 23 W

F-PLN

APPR	STAR	TRANS	FIX
RNV23-W	XAC 2B	BACON	TTE 178

CHK ITL, ITL /12 180kts, /7 160kts, TTE 178 (GA)



Missed App

L/H Turn **HME 177** OUTBD (PULL TRK)
(DCT UTIBO Radial In 357, MAN NAV)

RJBB(KIX) 17ft Gate 9

KE KIX 130.95 | UTC+9 | DCL -15min

ARR TL140 — SAEKI 170, RANDY 150

06R/L (059)	ATMUG B ATMUG A	BERRY ATVID	ILS Z
	After 2500ft L/G DN, After 1500ft L/D FLAP		
24L/R (239)	ATMUG M/C ATMUG D/E	MAYAH D/E Night	ILS Z

NAV — KIE 111.6 06L IKJ 108.7 06R IKD 108.1
24L IKN 110.7 24R IKW 108.5

06R : A7(5137'), A6(6938'), 24L : A8(5269'), A9(6976')
06L : B8(5160'), B6(6751'), 24R : B7(5318'), B9(6751')
24L Downslope

TAXI RTE 1(via J4), 2(via J3)

PA

DEP TA140 — SOUJA Tx

Walk Around 안에서는 정비사가 밖에서는 비밀번호

06R/L (059)	RINKU x - SOUJA Tx	ALT ATC(9000)
24L/R (239)		ALT ATC(9000)

TAXI RTE 1(via J4), 2(via J3)

- DEP 119.2

PA

IN
OUT

주황색 Landing Permit



RJAA(NRT) 135ft Gate 21-27

KE TOKYO 131.7 | UTC+9 | DCL -15min

PREFLT L/D은 북쪽 활주로, T/O는 남쪽 긴 활주로
TAXI GND Marking 잘 봐야함

ARR
TL140
HAKKA 330, YAGAN 240, LIVET 210, SWAMP 150

34R/L (338)
SWAMP E SWAMP T
ELGAR TYLER
ILS Z
L/G DN before ILS DME 14/12NM, **4NM L/D FLAP**

16L/R (158)
SWAMP G SWAMP N
GEMIN NORMA
ILS Z
ILS DME 4NM L/D FLAP

NAV
NRE 117.9 16L ITM 110.7 16R IKF 111.5
34L IYQ 111.9 34R ITJ 110.9

16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')
16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')

Arrival TAXI RTE in JEPP (No Numbering)

PA

DEP
TA140
ENPAR Tx

Walk Around 기계식 버튼 비밀 번호

16R/L (158)
TETRA x - ENPAR Tx
Check ALT
ALT ATC

34L/R (338)
ALT 7000(ATC)

Departure TAXI RTE 1, 2, 3, 4

- DEP 124.2

PA

IN
OUT
주황색 Landing Permit



RJFF(FUK) 30ft Gate 53-59

KE FUK 132.05 | UTC+9 | DCL -15min, Voice -5min

PREFLT 배풍 10kts까지 16사용, ILS 는 저시정에서만 사용

ARR TL140
PAVGA 13000
Hold W of IKE publish (Confirm FMS DATA)

16L/R (158)
ENTIX **RNP** LOC

RDR Vector Around IKE

34R/L (388)
R34RV KIGUS VIS (**RWY34**) ILS, RNP

NAV
DGC 114.5 (no A/P) 16L IFO 111.7
16R IFN 110.95 34R IFF 108.9 34L IFW 110.35

16L : C6(5475'), C7(6420'), 34R : C4(5124'), C3(6348')
16R : W7(5577'), W8(6696'), 34L : W4(5764'), W3(6466')

VIS 34 (Tailord Proc) : After IKE – RDR Vector Downwind – 1800ft – RWY Insight 1500ft – Before L/D CHK Complete before base **(Do not Extend Downwind due Terrain)**

DEP TA140
Consider W2, W9 Intersection DEP
PA

Walk Around

16R/L (158)
ALT ATC(10000)

HAKATA xx
L/H Turn RDR Vector

34L/R (338)
ALT ATC(10000)

EO 16L/R : DGC 158/20 R240

Caution 34L APCH HOLD LINE, RWSL in Operation
Initial CTC TWR, "Ready for departure"

- DEP 127.9

PA





RJGG(NGO) 12ft Gate 19-24

SWISSPORT OPER 132.05 UTC+9 DCL -15min

ARR TL140 SAMON 290, MARIA 130

36 (356)	CHESS(CARDS) SOUTH	PROBE	ILS Z
	After 1500ft L/D FLAP		
18 (176)	CHESS(CARD) NORTH	QUEST	ILS Z
	After 3000ft L/G DN & L/D FLAP		

NAV CBE 117.8 18 ICY 109.7 36 ICX 111.9

36 : A6(5213'), A7(6525'), A8(7837')
18 : A5(5393'), A4(6528'), A3(7841')

Caution STOP LINE, Yellow Ramp Line, VDGS!!

PA

DEP TA140 TANGO Tx

Walk Around 기계식 버튼 비밀 번호

36 (356)	OUMI x - TANGO Tx	ALT ATC(7000)
18 (176)		ALT ATC(7000)

APU Start 30min, Prepare Intersection T/O

- DEP 120.0

PA

IN
OUT

저녁식사후 하기 - 주황색 Landing Permit



RJCC(CTS) 70ft Gate 63-71

COM None | UTC+9 | NO DCL, Voice -5min

PREFLT 강설 주의, TAXI 불가시 Req Towing

ARR TL140
01R : IDEMI 150
19L : NAVER 170

01R/L (003)	**NAVER xxY** MUKAWA xxB	**YOTEI** Not YOSEI	ILS Y 3000 ILS Z 2000
19L/R (183)	**NAVER xxP** NAVER xxK	**PUPIG** KAORY	ILS Z **CAT III 19R only**

NAV CHE 116.9 01R ICH 110.75 19L ICM 109.35
01L ICN 110.9 19R ICS 111.5

01R : B4(5278'), B3(7047'), 19L : B8(5177'), B9(7119')
01L : A5(5538'), A4(6961'), 19R : A7(5390'), A8(6873')

Do not Cross 01L/19R After L/D (No TWY)
TAXI to Gate Via D(J) or G

PA

DEP TA140
Deicing at the Gate
APU 30분전

Walk Around 기계식 버튼 비밀 번호

01L/R (003)	DALBI xx SAVIT xx SOSHU xx	ALT ATC
19R/L (183)		ALT ATC

R/H Turn DCT HWE - Confirm R/H Turn ND

- DEP 124.7

PA



RJFK(KOJ) 891ft Gate 2,3,4

Japan Air KOJ 131.0 | UTC+9 | DCL -5min (0700-2200L) Monitor 121.8

PREFLT Caution Volcanic Ash Information Before DEP

ARR TL140 IKE FL260, KAZSA FL160 TERR RW16 Final
RW16 : Keep Manage NAV Until RWY Insight

34(337)	SIMAZ NORTH	SIMAZ	ILS Z
16(157)	OIDON	OIDON	RNP

NAV KGE 115.7 34 IKG 111.7

34 : T5(6387'), T6(7637',90도), 16 : T3(5990',90도), T2(7988',90도)
RW16 DownSlope

White Solid Line = Ramp Area
Follow Marshaller

PA

DEP TA140 Req Planned ALT in DCL : P340 in RMK

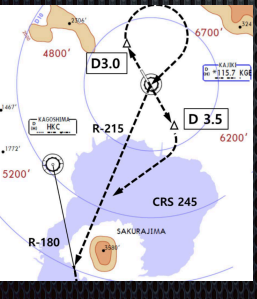
Walk Around 나갈때는 비번 들어올때는 불필요

34(337)	ATRUK x ASHIKITA Tx	ALT ATC(9000)
16(157)	Maintain RWY H/D	ALT ATC(9000)

EO 34 : KGE 337/3 R215, 16 : KGE 157/3.5 R215

- DEP 119.4

PA





ZGGG(CAN) 50ft Bay 147

China Southern 132.0 UTC+8 DCL -10-30min NO Read Back

PREFLT 여권 번호만 확인 제출X

ARR TL108 CAN CTL, APP Freq no Chart, Low GA ALT Short PTN AFT IAF, RW20 Windshear

02R01L (017)	OLPAB xx RNAV	GG611/GG612	ILS Z
20L19R (197)		GG511/GG513	ILS Z

SPD 8NM 180kts, 6NM 160kts

NAV
02R IDM 108.5 01L IGW 109.5 20L IXL 111.9
19R IKU 108.95 02L IBB 110.35 01R IOO 109.3
20R IAA 110.75 19L IPP 111.5 POU 114.1(no A/P)

02R:Y10(6227'),Y8(7263'), 20L:Y12(6217'),Y14(7240')
01L:N5(6453'),N4(7755'), 19R:N8(6423'),N9(7742')
Caution Cross RWY Intermediate TWY

STD Taxi RTE in JEPP(GREEN x), EAT 3/4 in JEPP
APU off Procedure, FLAP 0 for Refuel **PA**

DEP TA89 Consider Extra Fuel Delay/CRZ LVL

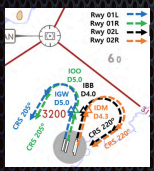
Walk Around 로툰다 엘베이용 비번 입력 V 누르고 엘베버튼 누름

02L01R (017)	BOVMA xx RNAV	ALT ATC(1200m)
20R19L (197)		ALT ATC(1200m)

EO 01L/R IGW/IOO D5.0 R205, 02L/R IBB/IDM D4.0/4.3 R220

STD TXY RTE in JEPP(Yellow X) Report RWY on DEP

- DEP 119.7(DEP RWY) **PA**





Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	**3,300 m**	**10,800 ft**
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZGSZ(SZX) 13ft Bay 301-319

SZX Dispatch 131.3 | UTC+8 | DCL AVAILABLE

PREFLT 여권만 제출

ARR TL108
SPD Restriction JEPP
Caution Freq CHG & RDR Vector (TERR!!)

1516R (156)	SAREX 3, 4 RNP	SZ462	RNP ILS Z
3334L (336)		NLG	RNP ILS Z

SPD 30NM 220kts, 8NM 180kts, 5NM 160kts

NAV 15 IQJ 111.3 16 ISZ 108.1 33 IMH 110.7
34 IBA 109.1 (no A/P VOR)

16R : H6(5449'), H8(6676'), 34L : H6(5413'), H5(6561')
15: C8(6725'), C9(8152'), 33 : C5(6548'), C4(8175')

Be carefull RWY 16L then TWY Q,R

CTC Apron Before Entering, MAX wingspan in JEPP
APU OFF but Remain

PA

DEP TA89
Consider Extra Fuel Delay/CRZ LVL
ATIS CTN V17, 18 Available -> SULAS 1 Dep

Walk Around 로툰다 중간 문 앞에서 통화버튼 누르고 ID + 얼굴 보여주면 열림

1516L (156)	IDUMA 1 RNP	ALT ATC(1200m)
3334R (336)	SULAS 1 RNP	ALT ATC(1200m)

EO 15 : IQJ 155/3.5, R205, 16 : ISZ 155/3.5, R205



DEP DCL (AUTO) Report DEP RWY

PA



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	**3,300 m**	**10,800 ft**
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZYTX(SHE) 198ft Bay 137-139

China Southern Dispatch 131.5 | UTC+8 | DCL -40min, Read Back Voice -10min

PREFLT 여권 제출, GD Sign

ARR TL118 CLR Limit TOSID Late Handoff to SHE Around TOSID - Present TRK,H/D - CTC SHE

06(056)	TOSID 62A, 61A	TX504	ILS Z
24(236)	TOSID 72A, 71A	TX603	ILS Z

NAV SEY 114.1 06 IPU 110.5 24 IKY 110.3

06 : D(6210'), C(7854'), 24 : J(6227'), K(7864') Prefer Exit

06 : HP09(08), 24 : HP04 Follow Me Car

Normally Remain Parking Brake SET!!

APU off Procedure (GND Air Cord & GPU)

PA

DEP TA98 A2, A8 Intersection T/O not Avail A330

ADT = CTOT See Enroute ATIS

Walk Around L1 비번없이 나가면 됨

06(056)	TOSID 61,62D	ALT ATC(900m)
24(236)	TOSID 71,72D	ALT ATC(900m)

"P/B follow GND Maintenance"

Be Careful "Hold Short CAT I Hold Line" Find Sign!!

Maintain Present TRK/HDG Join A588 (CRS 217)

Or Offset 3NM R/H of TRK

CTC APP without TWR Instruction

APP 121.225

PA



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZBAA(PEK) 116ft Bay 205-208

Air China Beijing 132.0 UTC+8 **DCL -40min, Voice -20min**
COBT/STD 15분 차이 CTC COM

PREFLT 여권 번호만 확인

ARR TL118 **RWY 01/19 Main L/D (RWY36L/18R)**

0136L (001)	**DUMAP xZ**	AA141	ILS Z 01 (Y 36L)
1924 (181)	**DUMAP xG**	AA261	ILS Z 19 (Y 18R)

SPD TMA Max 280, 8NM 180, 6NM 160

NAV **PEK 114.7 01 INJ 108.5 19 ISZ 108.9 36L IDK 111.7 18R ILG 110.3 36R IQU 111.55 18L IOR 109.3**

01 : Q5(5223'), Q6(7024'), 19 **3.2** :Q4(5298'),Q3(7103')
36L : P6(6276'), P7(7719'), 18R : P3(6223'), P2(7552')

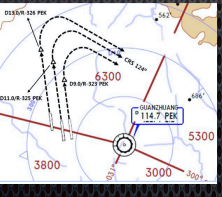
APU off Procedure (GND Air Cond' & GPU)
Standard TAXI RTE in Jeppesen Chart **PA**

DEP TA98 **RWY 36R/18L Main T/O(Intersection W2, W7)**
COBT on ATIS "Enroute", Bad Wx DOTRA SID

Walk Around

36R01 (001)	**MUGLO xX/Y** **Maintain RWY TRK**	ALT 2100m(1800m)
18L19 (181)	**MUGLO xG/J**	ALT ATC(1500m)

EO 36R : PEK 325/11, 36L : PEK 326/13
01 : PEK 323/9 -> R124



- APP 124.4 **PA**



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZHHH(WUH) 115ft Bay 331-335

COM None | UTC+8 | DCL -30-10min READBACK!!

PREFLT 여권 번호 확인 및 제출, GD Sign

ARR TL118
WUH CTL Rep HLxxxx
RDR CTC 60NM Prepare STAR CHG

0405R (046)	PONUD 01T PONUD 05T	HH310 HH357	ILS Z
1924 (246)	PONUD 02T	AVLEN	ILS Z

SPD Intercept Final to 8NM 180kts

NAV WHA 112.2 04 IHN 109.3 22 ITS 108.5 05R IMU 111.1
IQD 111.35 05L IWF 111.5 23R IUT 111.5

04 : B8(5557'), B9(6578'), 22 : B5(6738'), B4(7939')
05R : F10(5475'), F11(6597'), 23L : F7(5341'), F6(6545')

Follow Me Car TWY K/J
APU off Procedure (GND Air Cond' & GPU) **PA**

DEP TA98
Readback RWY, ALT to CLR
Report Ready for P/B to CLR

Walk Around 2층에서 엘베타고 내려가서 비번입력

0405L (046)	BIVIP 7S	ALT 4800m/ATC(900m)
2223R (246)	BIVIP 6S	ALT 4800mATC(900m)

Uncommanded Turn RNP DEP

- APP 119.575 **PA**



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZLXY(XIY) 1572ft Bay 556-561

Airport Oper Center 132.0 UTC+8 **DCL -30-10min READBACK!!**

PREFLT 여권 번호 확인 및 제출, Custom 작성, GD Sign

ARR TL118
SPD REST in REF Page
Req ILS APP instead of VIS APP

05LR (052)	LOVRA xZ/Y	XY906	RNAV ILS Z
23RL (232)	LOVRA xW/Y	XY801	RNAV ILS Z

SPD ZS/FNH 220kts, XY914/918/807/810 200kts, XY958/919/859/805 180kts, 5NM 160kts

NAV LCZ 109.0 05L IQU 108.55 23R IRA 111.7, 05R IXW 109.3 23L IAQ 111.1

05L : A3(6778'), A2(9032'), 23R : A6(5544'),A7(6512')
05R : D4(5613'), D3(7322'), 23L : D5(5646'),D6(7408')

Follow Me Car, CTC Apron Before Gate in **Taxi RTE in JEPP**
"Closing to xx TWY, apply to CHG to xx Freq

PA

DEP TA98
Apply NOTAM TO Performance
ADT = CTOT

Walk Around

05LR (052)	WJC xW	ALT ATC 1500m
23RL (232)	WJC xX	ALT ATC 1500m

PushBack GREEN/BLU used

- APP 119.9

PA



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

VHHH(HKG) 28ft Bay N5-N12

HAS FLT Dispatch 131.6 UTC+8

DCL -20min
5분 차이시 CTC Comm

PREFLT Deicing 유사 Delay시 시간 초과 가능성 있음

ARR TL110

ENPET 260 (1500-1659Z RWY 07R/25L X)
Terminal Tx RTE + STAR Chart
RWY 25R After TOPUN - APP Mode

07LRC (074)	ABBEY xA SIERA xA/C	LIMES	ILS
25RLC (254)	ABBEY xB SIERA xB/D	RIVMI	RNAV Tx ILS 25R ILS 25L

SPD Confirm Chart SPD vs ATC SPD (ATC First)

NAV SMT, 07L IZSL, 07R IZSR, 07C IZSC, 25R ITFR, 25L ITFL, 25C ITFC

07L : C7(5882'), C8(7194'), 25R : C6(5882'), C5(7211')
07R : J7(6916'), J8(7998'), 25L : J5(6916'), J4(8192')
07C : A7(5692'), A8(7208'), 25C : A6(5127'), A56673')

Gate In Solid line For Wide Body, APU OFF Proc

DEP TA90

SID + Terminal Tx RTE Chart
SID - Tx RTE Chart Many SPD Restriction

PA

Walk Around

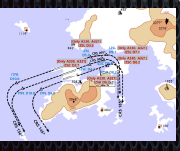
NADP 2

THR RED : 1000ft, ACCEL ALT : 1000ft
CLB SPD : GREEN DOT PRESEL
1000' NO LVR CLB - FLAP 0 - THR CLB - MAN SPD

07CRL (074)	DALOL xE/C/A (Z/Y/X RF LEG)	ALT 5000 NADP 1/2
25CLR (254)	DALOL xB/D/F	ALT 5000 NADP 2

EO

07L: IZSL074/3,R095, IZSL/9.1, 25R: ITFR254/10,R156
07C:IZSC074/2.7,R080, IZSC/6.2, 25C:ITFC254/10,R156
07R:IZSR074/6.2,R080,MAX200kt, 25L:ITFL254/6,R145



- DEP 123.8

PA



ZBTJ(TSN) 6ft Bay 105-107

Air China TSN 132.0 | UTC+8 | DCL -30min, Voice -10min READBACK!!

PREFLT

ARR TL118 — Missed APP 1970ft Initially

16LR (161)	DUMAP xK/R/H	TJ960 TJ919	ILS
34RL (341)	DUMAP xQ	TJ841	ILS

SPD 6NM 160kts (JEPP REF)

NAV TAJ 112.1 16R IJS 110.9 34L ICG 110.5
16L ICU 109.7 34R IKD 111.5

16L :W3(6269'),W2(9809'), 34R :W7(6443'),W8(7591')
16R :B4(5177'),B3(7191'), 34L :B5(5183'),B6(7201')

Follow Me Car on D, TAXI SPD Max 27kts

PA

DEP TA98 — Caution 600m Level Off - SPD Increase

Walk Around

16RL (161)	MUGLO xT	ALT ATC(600m,2000')
34LR (341)	MUGLO xW/R	ALT ATC(600m,2000')

16R : Do not pass A11

Confirm Parking Brake Release Before Push Back

- APP 119.27

PA



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

ZSQD(TAO) 30ft Bay 131-141

Comm None | UTC+8 | DCL AVAIL, Voice -10min READBACK!!

PREFLT 여권 번호 확인 및 제출, Custom Form

ARR TL118 AFT AGAVO - IKEKA - LATUX
(WEH 훈련시 RE-RTE됨)

3534 (350)	LAT 91A/01A	QD700	ILS Z
1716 (170)	LAT 81A/11A	QD305	ILS Z

SPD 6NM 160kts

NAV JDG 114.45 35 IQX 109.75 34 IIN 108.55
17 IQD 110.15 16 IDA 111.9

35 : S2(5255'), S4(6624'), 17 : S1(5282'), S3(6604')
34 : R2(5278'), R4(6650'), 16 : R1(5318'), R3(6706')

1st TWY GND, 2nd TWY Apron, CTC Apron
Caution HOT SPOT (HP01,HP02)
Follow Me Car on Lxx, APU off Procedure

PA

DEP TA98 Aft T/O - Heading 190, Join W209 -> DCT
LATUX CRS 148(328)

Walk Around Ramp Pass & Password 사용

3435 (350)	LAT 91D/01D	ALT 3000'(900m)
1617 (170)	LAT 81D/11D	ALT 3000'(900m)

Caution HOT SPOT (HP01, HP02)

PA

- APP 119.4 (RWY, INI ALT)



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

VTBS(BKK) 4ft Stand F,G 1-5

KE Bangkok 131.25 UTC+7 DCL -20min, Voice 133.8

PREFLT Check Payload

- TPE - HKG RTE

ARR TL130 SPD CTL via STAR Chart
20R/02L RNP Only

20R19 (295)	EASTE xxC	Select Tx (No Tx)	ILS Z 19/20L RNP 20R
02L01 (015)	EASTE xxD	Select Tx (No Tx)	ILS Z 01/02R RNP 02L

NAV SVB 111.4 19 ISEN 110.5 02R ISWS 109.1
20L ISWN 109.5 01 ISES 110.1

SPD 10M' 250, 8NM 180, 5NM 150-160

19 : B8(5567'), B10(6965'), 01 : B7(5964'), B5(7962')
20L : E9(5052'), E13(7139'), 02R : E12(4872'), E7(6958')
20R : F7(6994'), F9(8471'), 02L : F6(5725'), F4(7496')
HIRO, 01/19, 02L/20R No Groov

AFT LD CONF 1, STD Taxi Route, APU OFF Procedure

PA

DEP TA110 A-CDM REQ P/B +-5min of TSAT
TSAT/CTOT Inform to GND Staff

20L19(295)	LIPLI xJ, G/E	ALT 6000
02R01(015)	LIPLI xK, F/H	ALT 6000

APU Start within 10min, STD TAXI Route
19R Do not Pass E1, D2 01/19, 02L/20R No Grooved

- DEP 119.25 (AUTO)

PA

IN	Fast Track, CAPT GD, Sign 여러장
OUT	New CREW Line (에스컬레이터 밑), 여권, ID, GD , Sign 여러장



RPLL(MNL) 75ft Gate 5,6

PAGSS Oper 131.0 | UTC+8 | No DATIS
DCL -5min 125.1 Voice

PREFLT 입국 세관 신고서 (https://etravel.gov.ph/)

- Report ETA BISIG

ARR TL130 3 STAR Diff TRK NM (7>5>3) & IAF "RPHI"
GPS Interference Gate Info 11L1(L1만 사용)

06(063)	POLIO, NABAL 7,5,3 R/P	GONDO (DAGAT)	ILS RNP
24(243)	DCT MIA RDR Vector	MUTAN (MEDAM)	ILS RNP

NAV MIA 114.4 06 IML 109.1 24 IMA 109.9

SPD TMA 250, 20NM 210, 10NM 180, FAF 150

06 : E2(7555'), R1(8221'), 24 : E4(6725'), R5(7755')
Unable advise ATC, Do not Confuse R2, E2, RWY31

06 : Caution Vacate & RW31, 24 : TWR to APRON CTC
In Apron S4,5 Small Marking, Ask Chock in Time

PA

DEP TA110 Main RWY H/D Climb 7000, CLR/T/O

06(063)	CAB xx R/A	ALT 12000/ATC
24(243)	CAB xx P/B	ALT 9000/ATC

EO 06 : MIA 063/2, MIA R250

Log on CDPLC RPHI Before DEP.
Req ENG Startup to GND -> Req P/b to Apron



- DEP 124.4
- Report ETA LEBIX

PA

IN 직원 IN/OUT GD, INFO 줌, IN GD 및 QR 제시

OUT 받은 OUT GD 제출



WSSS(SIN) 22ft Gate G18-21

SATS FLT OPER 131.22 UTC+8 DCL -20min, Voice -5min

PREFLT 신고서 작성 (https://eservices.ica.gov.sg/sgarrivalcard/)

ETA KABAM 119.3 – ETA MIGUG - MAVRA 8942/5655 – MIGUG Rep
MESOX 120.7 - Rep MAPNO (RPHI – VVHM CPDLC Tx Manual)

ARR TL130
CPDLC WSJC -10min FIR
SPD CTL via STAR Chart, No IAF RDR Vector

02LC(023)	MABAL/KARTO xA	No Tx	ILS
20RC(203)	MABAL/KARTO xB	No Tx	ILS

RNP 02R/20L : L/D CHK -> LUVUL/OBGIS -> Arm APPR

NAV
VTK 116.5 02L ICW 110.9 02C ICE 108.3
20R ICH 108.9 20C ICC 109.7 02R ICX 110.5

SPD 8NM 180kts, 4NM 150kts

02L : W5(6519'), W4(7916'), 20R : W6(5498'), W7(6423')
02C : T6(6781'), T5 (8467'), 20C : T7 (6420'), T8 (7910')
02R : A6(6738'), A5 (7995'), 20L : A7 (6250'), A8 (7395')

HIRO Unable ATC, Exit TWY 2 way but Same Name
VTK Out of A/P, TWY R/S Max 20kts, AFT LD CONF 1

PA

DEP TA110
Req P/B +-5min TSAT
CPDLC WSJC Before DEP

Walk Around L1 과 L2 브리지 중간에 Security가 서있는 곳으로 가면됨. 다녀와서는 몸 수색함

20CR(203)	TOMAN xx CHANGI xD	ALT 3000 SPD 230 Until 4000
02CL(023)	TOMAN xx CHANGI xC	ALT 3000 SPD 230 Until 4000

PA

Standard P/B, TWY R/S Max 20kts
Cross LUSMO/AKMON xxx, T/O Before xxx
RWY 20C/R : CHK Min Climb Grad by NAVBLUE

DEP 120.3 - GUTUP (SIN 6556/8942) - HCM AKMON 133.05
- UDOSI 120.7 - ARESI 8942/5655 Unable AKOTA 119.3

IN
OUT
기계에 여권 인식하는 방식, Crew Line 따로 있음



VVDN(DAD) 30ft Bay 14-20

Comm None | UTC+7 | GND By Voice

SANYA DB Report 122.6

ARR **TL100**

C/S A330 Heavy!!

SPD CTL via STAR, VOR 17 Available

35LR(353)

BUNTA xM/N/L/K

L/K : RNAV Arr Chart | TAMLA | ILS V

Long STAR Primary

17RL(173) **3.5**

BUNTA xD | DAN | VOR Z

BUNTA xE | D057O | VOR Y

NAV DAN 114.4 35R IDR 111.5 35L DAD 110.5

SPD FL100 Max 250, 12NM Max 200, 5NM Max 160

35L : G4(6702'), G6(9803'), 17R : G2(7089'), G1(9776')

35R : E4(7198'), E6(10275'), 17L : E2(8044'), E1(10754')

NO RAPID EXIT TWY, 90 Degree TURN, AFT LD CONF 1

Follow me Car on E, E2,3,4 Narrow Over-steering

TWY D 사용 금지, 소형기 on D시만 E 사용가능

Gate IN From TWY E Only, 5m 1kts Ready for STOP

PA

DEP **TA90**

Consider Extra Fuel Due FLT ALT

ATS Plan HardCopy 제공

35RL(353)

BUNTA xG/C

RNAV Deps | ALT 8000(ATC)

17LR(173)

BUNTA xA | ALT ATC

EO

35L/R : DAN 353/5, R060

17L/R : DAN/173/4.5, R130

Complete P/B – ENG Start, Follow me Car

35R CAT I Hold Point Before E1, 17L Hold Short 가까움

TWY D 사용 금지, 소형기 on D시만 E 사용가능



PA

IN 직원이 주는 GD 제출

OUT GD 불필요



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

VVTS(SGN) 33ft Bay 9-16

Comm None | UTC+7 | Voice -15min, Del 121.8

MNL 119.3 - Monitor MNL RDO 8942(5655) - At MIGUG HCM 120.7

ARR TL100	CPDLC VVHM		
25RL(250)	PTH xD DOVIN xD/E	SULUK BOMSU	ILS W
07LR(070)	PTH xM DOVIN xP/Q	SAMDU	ILS W (VOR 07L)
NAV	TSH 116.8 25R HCM 110.5 07R ITS 111.7 25L SGN 108.3		
SPD	25NM 190-200, FINAL 160-180, 5NM 150-160		

25R : P4(6158'), P5(6991'), 07L : P3(6266'), P2(8907')
07R DIS :S5(6574',110도) S3(7598') then V(Heavy)
25L : S7(6824'), S8(9671') Do not Confuse TWY S/S8
After Vacate then TWY V

AFT LD CONF 1, Follow Me Car Service in Ramp
(Caution Sudden STOPBAR L/T)
Sensitive VDGS!!! (0.5m이내, 2m STOP시 바로 정지)

PA

DEP TA90	Req RWY due to Performance Aft T/O Sudden Low ALT Maintain 2000ft	
Walk Around	직원이 문열어주고 기다림	
25LR(250)	PATMA xA/M/P PTH xC	ALT FL110 ALT 9000
07LR(070)	PATMA xF	ALT ATC

P/B within TSAT +- 5min, RWY CHG After TAXI
Caution Sudden STOPBAR L/T, Follow Car Service
1,2,3T is Stand name, Caution S1, P1 for Lineup

DEP 125.5

PA

IN	직원이 주는 GD 제출
OUT	호텔 CHKOUT시 GD 1장(Sign) Imm 제출, 신발 off



VTCC(CNX) 1036ft Bay 5-7

CNX Traffic 131.5 | UTC+7 | Voice -5min, GND

ARR TL130	BKK to CNX APP 80NM, FL160 - Rep PSN, ALT, SQ Caution TERR

36(001)	MONLO xA	PUKAT	ILS Z
18(181)	MONLO xB	PILEX	RNP

NAV	CMA 116.9 36 ICMA 109.9
SPD	20-25 TRK NM 220, IF 180, FAF 150-160

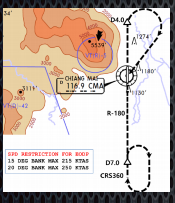
36 : E(6558', 90도), D(7483')
18 DIS : F(6853', 90도) , G(9199', 90도)
Narrow RWY, No TWY CL L/T, Yellow Center Line
Difference VDGS, AFT LD CONF 1

PA

DEP TA110	AFT T/O L/H TURN DCT MONLO then R-207 CHK CLB Gradient, RECMD FLEX Due EGT Limit

36(001)	MONLO xN	ALT 11000
18(181)	MONLO xS	ALT 6000

EO	36 : CMA 001/4, R180	18 : CMA 181/7



DEP 129.6

PA

IN	BKK 처럼 CAPT GD Sign 5번, Gate 7번 앞 버스 대기
OUT	GD Sign 5번



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

WADD(DPS) 14ft Stand A15-25

Gapura Oper Bali 131.95 UTC+8 Voice -20min, TWR/GND

PREFLT QR code 작성 (https://allindonesia.imigrasi.go.id)

CPDLC - POTIB 10-25 min RPHI - MAMOK 15-30 min WAAF

ARR TL130 RW27 Before DOBMA(EMTAS) FL300 for CTC Bali 119.7 (Prevent Late DES) Late DES due TERR

09(087)	DOBMA/EMTAS xC	KHUTA	RNP
27(267)	DOBMA/EMTAS xD	BENOA	ILS

NAV BLI 116.2 27 IDPS 110.3

SPD 10NM 180kts, 4NM 150kts

09 : N3(6817', 90도), N2(8448', 90도)

27 : N6(7513', Rapid) , N7(9645', 90도)

Narrow RWY, No RWY/TWY CL L/T, Yellow C/L Only

VDGS but Marshaller, GND REG No, After LD CONF 1

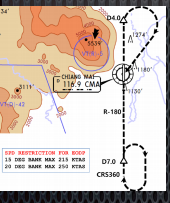
PA

DEP TA110 +-15min Req P/B, Rep Bay, Resist No, POB CPDLC Before DEP WAAF

09(087)	DOBMA/EMTAS xA	ALT 9000
27(267)	DOBMA/EMTAS xB	ALT FL240(ATC)

EO 09 : BLI 087/6, R130

Caution Intermediate HP W/E



DEP 119.7

CPDLC

- Before DEP WAAF - GORAI 10-25min RPHI
- GURAG RJJJ

PA

IN	도착후 직원이 주는 입/출국 카드 작성, GD 불요, QR 제시
OUT	KE 카운터 앞 Dispatch 브리핑 후 인솔, Imm 이후 STAFF Only로



VDTI(KTI) 22ft Bay 107-123

Techo DIS 129.0 UTC+7 Voice P/B -10min, GND

CPDLC- POTIB 10-25min RPHI, MONBO 15–45min VVHM
ETA KABAM(RTA) – ETA MIGUG - Report MAVRA, Report MIGUG
At MIGUG CTC HCM 120.7

ARR TL ATIS	Do not Confuse ANAT, ANAB TCAS RA Des Rate!!		
05(048)	SUSEL/ELNAK xD	NULPI/IKOMO	ILS Z
23(228)	SUSEL/ELNAK xC	AKPEG/LELUN	ILS Z
NAV	TIA 111.0	23 IKTI 110.1	05 IKTA 108.75

05 : A5(6562'), A6(7677'), 23 : A4(6562'), A3(7677')
Rapid Exit, Non-Grooved RWY

Caution Complex Lead Line into MARS

PA

DEP TA100	RWY 23 Do not Overshoot B17	
05(048)	ELNAK x	ALT ATC (6000)
23(228)	SUSEL x	ALT ATC (6000)

APP 123.8 (AUTO)
CPDLC
- AKOTA 15-45min VVHM - POTIB 10-25min RPHI

ETA ARESI - MNL RDO 8942/5655 At ARESI
MNL 119.3 At -10NM AKOTA - ETA POTIB

PA

IN	GD 2장, 입국, 출국 각각 사용
OUT	출국장 맨 왼쪽 CREW LINE



RCTP(TPE) 108ft Bay A1-9

Dynasty Oer 131.3 UTC+8 DCL Avail, Voice -5min

ARR TL 130	COPRA RL200		
05LR(054)	BAKER xA	JAMMY	ILS
23RL(234)	BAKER xB	AUGUR	ILS
SPD	FL250-130 280, FL130-10M' Max 280 Ref APP		
NAV	no VOR 05L ITIA 111.1 23R ITYA 109.3 05R ICKS 110.7 23L ICJN 111.9		

05L : N7(5787'), N8(7411'), 23R : N4(6656'), N3(9340')
05R DIS :S6(5419'),S7(7244'), 23L DIS :S5(5442'),S4(7470')

Center Green, Deviation by Arrow, Distance Yellow



PA

DEP TA110	Be Ready Intersection T/O In DCL "MNTN A030" -> 3000ft	
05RL(054)	PIANO xA/C	ALT ATC
23LR(234)	PIANO xD/B	ALT ATC

"DCT PIANO then L3 RNAV Transition"

PA



VYYY(RGN) 110ft Stand 6,7

Oscar Papa 131.8 | UTC+6:30 | Voice -15min, GND

PREFLT 승무원 2 SET 운영, 탑승 정비사

NAR(No Alternate Airport) not apply due Single RWY

ARR BGO(BAGOO), HGU(HLEGU) ILS Z GNSS Req
TL 075 GPS Interference -> RDR Vector + ILS Y 21

21(214)	**GPS OUT** **RELAP xA**	RDR Vector PAKSU	ILS Y ILS Z
03(034)	**BAGOO xB**	KOSDA	RNP

NAV no A/P VOR BGO HGU 21 IYGN 109.9

21 : A3(8152'), R1(8415'), 03 : A5(7972'), A6(11069')
RWY 21 Upslope(0.6%), Exit End of RWY or By ATC

Caution WingTip (Watch out Safety Line)
Ramp Area Dark(Consider LD L/T)

PA

DEP
TA60
GPS 1+2 Fail MEL34-50-08F, 34-50-09B(RTG)
No RNP RTE, ADS-B No RVSM Req Equip -> RADIONAV Updaate within 1NM No Problem

SEL **RW21**	**RDR Vector** (VISAM xA/C)	**SEL HD 214** ALT ATC SID GNSS Req
SEL **RW21**	**RDR Vector** (BAGOO xA)	**SEL HD 034** ALT ATC SID GNSS Req

RWY 03 Taxi Upslope(more PWR)

RGN RDR 128.75
- After BOMAS GPS Back to NOR
- Sometimes Spoofing

PA



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

WMKK(KUL) 69ft Stand C2,4,12

Layang Sat. 132.9 UTC+8 Voice TSAT -10min

ETA KABAM 119.3 – ETA MIGUG MAVRA 8942/5655 – MIGUG Rep MESOX 120.7 - Rep MAPNO (RPHI – VVHM – WSJC - WMFC)

ARR	ATC Req POB	
TL130	Long TRK Mile STAR Expect Short Cut	
33R33(326)	SAROX xG/H	EGURI ILS
14L15(146)	+ Initial APP Chart	MESUP
SPD	20NM 220, 15NM 185, 10-5NM 160	
NAV	VKL 116.1 33L IWL 111.9 33 IWM 111.5 14R IWR 110.7 15 IWK 110.1	

32R : A5(6938'), A4(8343'), 14L : A6(6922'), A7(8425')
33: Y5(6948'), Y4(8425'), 15 : Y6(6948'), Y7(8415')

RWY 32L via C5 Caution D5,6
Wingtip CLR P1-3, Q5-7, Y1-8 ATC Req HLxx/LD Time
Standard TWY RTE, AFT LD CONF 1

PA

DEP	Req P/B +-5min of TSAT	
TA110	CTOT from Staff	
32L33(326)	KIMAT xE/F	ALT ATC(6000)
14R(146)	KIMAT xB RNP 1600-2100Z	ALT ATC(6000)

Standard TAXI RTE, P/B Exit S8 -> Taxi out S8
Cross DUDIS xxx, T/O Before xxx

DEP 135.25(Before 2000ft)

GUTUP 6556/8942 – AKMON 133.05 - UDOSI 120.7
ARESI 8942/5655 Unable AKOTA 119.3

PA

IN	브리지 밖 직원 GD, 2300L 이전 Aerotrain, 이후 BUS(C5 Sign)
OUT	호텔 GD 수령, Aerotrain to Gate



VVCR(CXR) 46ft Bay 36

Comm None | UTC+7 | Voice TWR 118.2

ETA KABAM 119.3 – ETA MIGUG MAVRA 8942/5655 – MIGUG Rep
MESOX 120.7 - Rep MAPNO (RPHI – VVHM)

ARR TL100
Wx, RWY from HCM, CHK LDW due to PCR
RWY20 Max Tail wind 15kts
CAAV STAR, APP not Authorized

20LR (200)	COTUN, BANKE, HUNTA, NHATA xx	CRxx AFT IAF STEEP DES(3.8)	ILS Y 20L RNP 20R
02RL (020)			ILS X/Z

NAV CRA 116.5 20L ICA 110.3 02R ICR 111.9 02L ICR 110.7

20L : G3(6735'), G1(9603'), 02R : G5(6528'), G7(9662')
20R : W4(5971'), W3(7680'), 02L : W5(5606'), W6(7345')

FollowMe Car Service, Sensitive VDGS Caution

PA

DEP TA90
CHK TOW due to PCR
Follow Restriction due to Military Traffic

02LR(020)	NiHOA xA	ALT 7000
20RL(200)	NiHOA xB	ALT ATC(FL100) SPD 250

EO 02 : CRA 020/3, R090, 20 : CRA 200/6, R150

TWY Y5 only below wingspan 36m/118ft

DEP 127.9
HKG RTE

PA

IN
OUT

여권 제시 특이사항 없음



VVPQ(PQC) 23ft Bay 7,8,11,13A

Comm None UTC+7 Voice GND 121.92

HCM 120.9 (STAR,RWY,Wx) – No APP Freq

At OSOTA PQC TWR 118.6

ARR CPDLC VVHM

TL100 STAR RWY Wx from PHN or HCM(120.9)

28(276)	OSOTA xE TRUMI xC	PQ328 TUNPO Tx, PQ428	ILS Z
10(096)	OSOTA xF TRUMI xB	PQ310 TUNPO Tx, PQ210	ILS Z

NAV PQU 113.0 28 IPH 108.7 10 IPQ 108.1

28 : S6(6758'), S8(9662'), 10 : S4(6761'), S2(9665')

RWY 28 PAPI R/H not Accurate

AFT LD CONF 1, Stand 13A only Widebody

(Caution Sudden STOPBAR L/T)

Sensitie VDGS!!! (0.5m이내, 2m STOP시 바로 정지)

PA

DEP Req RWY Due to Performance

TA90 AFT T/O Sudden L/O 2000ft

28(276)	OSOTA xC	ALT 6000 SPD 250
10(096)	OSOTA xD	ALT ATC

LINE UP via RWY 28 S2/S1 or RWY 10 S8/S9

PA

TWR 118.6

- Before FL100 PNH 132.4/124.7 or HCM 120.9

IN	GD 2장, IMM 1장, 세관 1장(세관은 안받음)
OUT	KE 카운터 GD 2장, IMM 1장, X-ray 1장



VVNB(HAN) 43ft Bay 15-21

Comm None | UTC+7 | Voice -15min GND

SANYA BD Report 122.6

ARR TL100 — TEBAK RL240, SPD CTL via STAR Chart
GS Fluctuation

11LR (109)	NAKHA xH/E/F/G J/K : RNAV ARR	VILOX NB362	ILS X
29RL (289)	NAKHA xL/M/N/P RNAV ARR	NB462 NB452	ILS X

SPD — 25NM 190-200, 10NM 180-160, 5NM 15-160
Max RWY 40, TWY 30, Apron 15, Turn 10

NAV — NOB 116.1 11L NB 110.5 29R INA 109.3
11R NBA 108.3 29L INB 111.9

11L : P7(6929'), P8(8366,90도), 29R : P4(6361'), EOR
11R : S7(7555'), S8(10377'), 29L : S4(7732'), S3(10213')

AFT LD CONF 1, Follow me Car on S
5m 1kts Ready for STOP

PA

DEP TA90 — Consider Extra Fuel Due CRZ Level
TEBAK FL250

11RL(109)	NAKHA xA CHK NB110	ALT ATC
29LR(289)	NAKHA xC	ALT ATC

STD TAXI RTE - "DEP 1B"
TWY S1,3,5,6,8 항공기 있으면 TWY S 금지

DEP 125.1
HKG RTE

PA

IN	GD 불요, IMM 좌측 CREW LINE
OUT	DEP B(둘중 오른쪽)



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

YBBN(BNE) 15ft Stand 78-85

Menzies BNE 132.55 UTC+10 DCL -5min Voice

PREFLT CREW Declaration 작성

CPDLC & PSN Report : RJJJ – KZAK(FIR) –AYPM(FIR)-YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

FIR PSN Report Caution Wx DEV Sequence

ARR TL110 Check ALT A/P VBCG L/D WT
SPD Restriction in Chart + PRM

01L/R(016)	**TANIK**/GLENN	ILS
SMOKA xA		
19R/L(196)	**OSOKO**/ESTSO	ILS

NAV BN 113.2 01L IBN 111.5 19R IBE 109.3
01R IBA 109.5 19L IBS 110.1

SPD 5NM 150-160kts

01L : T6(6243'), T4(7598'), 19R : T9(6246'), T11(7595')
01R : A4(7043'), A3(8579'), 19L : A6(5620'), A7(8477')
Prefer Exit Unable ATC, min REV TH, T5, T10 No Exit

AFT LD CONF 1, CTC Apron Before Entering
Report Cockin Time Caution Hotspot RWY 31

PA

DEP TA100 CPDLC YBBB on GND 45min/Passing 10M'
"CNL SID, TRK EXT CL, xx Degree" -> Main RWY TRK

01L/R(016)	BIXAD x 01R Diff Chart	ALT ATC(6000)
19R/L(196)	BNE x RDR SID with H/D	ALT ATC(6000)

PushBack Point C,D,E , TWY A for T/O
Do Not PASS B1 for 19L

- DEP 133.45

PA

IN	로툰다 나온 직후 KIOSK 사용(한산함)
OUT	DEP 입구옆 직원라인 진입

EDTO
Next Page



EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning
(Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)
RVSM CHK : **CAPT/FO 20ft, PILOT/FE 75ft**
(PFD/ISIS 60ft, ST-BY 300ft)
NAV DATA Input : **EEP, ETP1, ETP2, EXP**
HF SELCAL CHK : Jeppesen - ENT DATA Pacific
SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : **CAPT/FO 200ft**

BEFORE EEP (Entry Point, ERA 기준)

60min(Theshold) : A330 420NM
One Eng : CRZ SPD 310kts, 180min 1235NM
FIX 1 : **EEP**, FIX 2 : **ETP1**
FMS ALT A/P SET : **DATA – ETP Page(Nearest AP)**
EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**
Review Contingency Procedure
- Drift Down 30도이상, 5NM, FL290이하, +-500ft
- Wx Dev 5NM 이상, +-300ft

EDTO Segment

Apply Actual Wx for Actual Divert
TERR Escape RTE Papua New Guinea

COMM (CPDLC)

RJJJ – KZAK(FIR) – AYPM(FIR) – YBBB(FIR)
(Cross GUM airspace 250NM 118.7, SQ2100)

Diversion



- Enroute RNP downgrade
- Weather min ERA AP
- LAND ASAP by ECAM, QRH
- Only 1 GEN(IDG, APU, EMER) + Multi Fail
- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)
- Increased Fuel Consumption

ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET
EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**
Last ETP(Critical Point) Fuel less then PLAN **– Continue by PIC**

EXP (Exit Point)

YSSY(SYD) 21ft Stand 78-85

Menzies SYD 129.6 UTC+10/11

Voice CLR 133.8/GND 121.7
No DATIS

PREFLT 여권, Custom Form

CPDLC & PSN Report : RJJJ – KZAK(FIR) –AYPM(FIR)-YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

FIR PSN Report Caution Wx DEV Sequence

ARR
TL110

RW34L/16R on REQ
CHK SEMAP, Curfew 2300-0600L
ILS PRM, NADP Proc Review

16LR(155)	BOREE xA (xP for 16R)	URDEN	ILS
34RL(335)		SOSIJ	ILS

NAV SY 112.1 34L ISN 110.1 16R IKS 109.5
34R IKN 109.3 16L ISS 110.9

SPD 10NM 185-160, 5NM 160-150

34R : U1(5515'), L(7378'), 34L : A3(6207'), A2(6620')
16LDIS:T4(5718'),T6(6971'), 16R : A3(6210'),A4(6561')
Prefer Exit Unable ATC, min REV TH, NO LASHO

SY 20NM Below 9000ft, Intl x, Dom x - TWY
Same Spot no. International/Domestic

PA

DEP
TA100

CPDLC YMMM on GND 45min/Passing 10M'
CASA no Approval KAL (for T/O min)

16RL(155)	GROOK x RIC tx (SYD x RDR)	ALT ATC SPD 270
34LR(335)	RIC tx (SYD x RDR)	ALT ATC

CHK Special Requirement SID Chart

- DEP 123.0

PA

IN

OUT

EDTO

Next Page



EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning
(Ops Pecs C055) -> EDTO ERA Only(ERA no Wx)
RVSM CHK : **CAPT/FO 20ft, PILOT/FE 75ft**
(PFD/ISIS 60ft, ST-BY 300ft)
NAV DATA Input : **EEP, ETP1, ETP2, EXP**
HF SELCAL CHK : Jeppesen - ENT DATA Pacific
SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : **CAPT/FO 200ft**

BEFORE EEP (Entry Point, ERA 기준)

60min(Theshold) : A330 420NM
One Eng : CRZ SPD 310kts, 180min 1235NM
FIX 1 : **EEP**, FIX 2 : **ETP1**
FMS ALT A/P SET : **DATA – ETP Page(Nearest AP)**
EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**
Review Contingency Procedure
- Drift Down 30도이상, 5NM, FL290이하, +-500ft
- Wx Dev 5NM 이상, +-300ft

EDTO Segment

Apply Actual Wx for Actual Divert
TERR Escape RTE Papua New Guinea

COMM (CPDLC)

RJJJ – KZAK(FIR) – AYPM(FIR) – YBBB(FIR)
(Cross GUM airspace 250NM 118.7, SQ2100)

Diversion



- Enroute RNP downgrade
- Weather min ERA AP
- LAND ASAP by ECAM, QRH
- Only 1 GEN(IDG, APU, EMER) + Multi Fail
- Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)
- Increased Fuel Consumption

ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET
EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**
Last ETP(Critical Point) Fuel less then PLAN **– Continue by PIC**

EXP (Exit Point)

VNKT(KTM) 4395ft Spot 1-4

Nepal Airline Corp 131.6 UTC+5+45

No DATIS
Boarding ATC CLR

CPDLC VYYF(Yangon), VECF(Kolkata)

Kolkata -10min (125.77/133.75) - Dhaka -10min ADC No. (125.7) - KTM -10min (126.5) C/S, A333, PSN, FL, FIR Time RTE, L/D type "Req Entering CLR"- Report Kolkatka Establish with KTM

ARR
TL150

Curfew 1915-0015Z
Night : Only RNP Z 02(AR) Comm Policy

02(022) ROMEO xR RATAN RNP Z 02 / VOR 02

SETUP

Nav : KTM 105&288, Fix : KTM /4, DANFE
FMS : KTM530/D9.0 (Vapp)

NAV KTM 113.2 02 IKT 108.1

02 : B(6814'), A(8395'), 20 : E(6387', 90°), 180 Back
High FE + Upslope = 50' Flare (No 40', 30')

RAMP SLIPPERY NO Release Brake **PA**

DEP
TA135

Req ADC & FIC, P/B with POB, Intersection A, E
If Wx is Severe, Stand-by at GATE

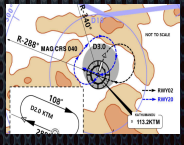
20(202) IGRIS xA ALT ATC

02(022) IGRIS xB ALT ATC

EO 20 : KTM /4, R340, R288 02: KTM /3, R288
Maintain CONF Until 10500ft, 8000ft LVR MCT(10min)

RW20 Takeoff at Painted Threshold
D2.5 KTM/KTM Turn Even Below 400ft (Max 180kts)

DEP 120.6 CPDLC VECF, VYYF
- Dhaka -10min ADC No. (125.7)- Kolkata -10min (125.77/133.75)
Radio Problem
- -80NM KTM FIR to Kolkata FIR +30NM
KMG 125.75(15-20min FIR)



PA

IN

OUT

RNP AR Next Page



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

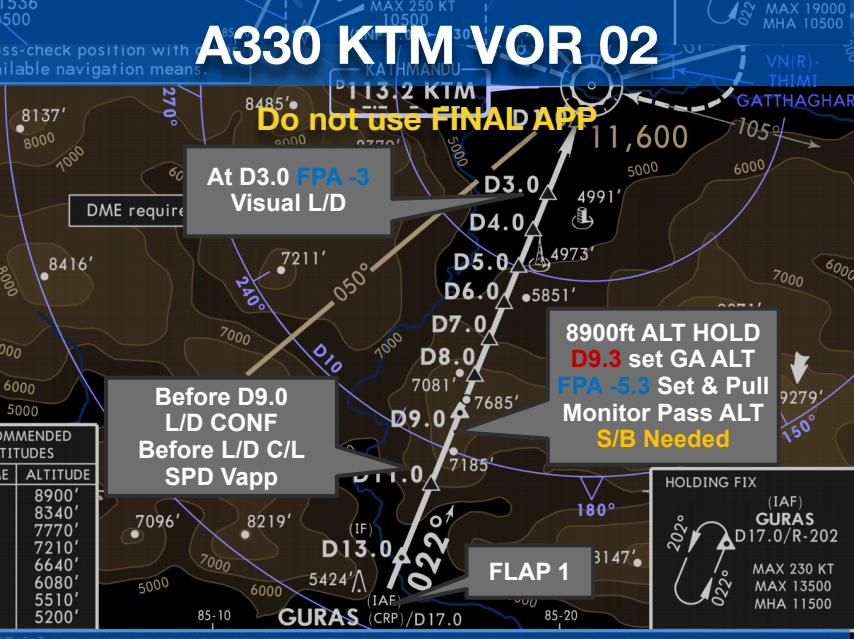
ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

A330 KTM RNP Z 02(AR)



A330 KTM VOR 02



SYS Fail / GA (Dual-GA, Single-GA or CONT)

if Follow GA RTE (Acc HT 10500ft)
- Reach 10500ft – **TH CLB - Acc** – Flaps up – Req DANFE

Dual FMGC (NO GA RTE) (Acc HT 10500ft)
- Man GA, 15°, 10-20kts ADD **(At RW02 R/H Turn)**
- PM **NAV B/UP** both **MCDU Menu** – TRK/FRA – Follow TRK (A/P ON)
- **if No GA RTE Follow VOR GA Procedure**
- Reach 10500ft – **TH CLB - Acc** - Flaps-up – Req DANFE

VOR GA Procedure (Acc HT 10500ft)
At RW02 R/H Turn **(Max 185kts)** – 4 DME Arc(IN 5 DME)
After RW02 Track 288 OUTBD Until Reach 10500ft

ZMCK(UBN) 4485ft Spot 2,4,6

Comm None | UTC+8 | No DATIS | Voice TWR 129.0

5min Before POLHO

Boundary Report to UBN 126.5

ARR TL148

HI FE, TERR/Restrict Area

CTC UBT 5min Before FIR(POLHO, INTIK)

HI FE&Sink Rate due TAS

29(290)	MENUG xW	ZONAM	ILS X
11(110)	MENUG xV	CK113	RNP(Req)
NAV	SER 116.9	29 CDA 109.9	

29 :C(5761'), B(8241'),END RWY, 11 : C(5761'),D(8231')

Gate 2 A330 Must Follow Dashed Line

PA

DEP TA128

Req RWY 11 DEP Available

Walk Around 정비사에게 비밀번호 요청

29(290)	MENUG xA/C	ALT ATC
11(110)	MENUG xB/D	ALT ATC

EO

29 : SER 290/4, R343, SER R133, SER 290/10, R260

11 : SER 110/1.5, SER R313, SER 290/10, R260

TWY F very Narrow, TWY B,D Opposite Turn Prohibit

PA



DEP 120.0 UBN 126.0 - 126.5

Boundary Report 15-20min



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	**4,500 m**	**14,800 ft**
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

OMDB(DXB) 62ft Stand C45-63

DNATA Oper 131.6 UTC+4 **DCL Avail, Voice -10min**

S RTE : CHILA Report YANGON, DAKHA, KOLKATA

Boundary Report - PURPA(W112) : islamabad Before 15min

- APELO(B505) : Karachi Before 15min- DERBO(G452) : Air Defense RDR Before10min - over IRAN : Air Defense RDR (127.8/135.1)

PSN+FL+DEP+DEST (Separate Tehran CTL)

ARR
TL150

Arr : C/S,A333, Passing FL -> Director : C/S Only, SPD Restriction in JEPP

12LR(119)	**PUVAL, IMPED xB**	REREK	ILS
30LR(299)	**PUVAL, IMPED xC**	ULDOT	ILS

NAV

12L IDBL 110.1 30L IDBW 111.3
12R IDBE 109.5 30R IDBR 110.9

12L : **M7A(6135'),** M9(7253'), 30L : **K9(6811'),** K8(8041')

Late L/D CLR, Assigned Rapid Exit TWY

TWY J,J1,J2,J3,U,W,Y,Z Max 15kts, VDGS 2kts
Caution TWY Holding Point, No Wing Walker

PA

DEP
TA130

Full Length to ATC

30R(299)	**DAVMO, IVURO xF**	ALT 4000 **SPD 220**
12R(119)	**DAVMO, IVURO xG**	ALT 4000 **SPD 220**

DEP 124.675 C/S, Passing LVL, Unable SPD

CTC Air Defense RDR 127.8/135.1 Before 10min Tehran FIR(GABKO FL190) - MUM HF : 6661/8879/5658

S RTE : CHILA Report YANGON, DAKHA, KOLKATA

Boundary Report - KMG 125.75(15-20min FIR)

PA

IN	GD 2장 - IMM 1장, OUT시 1장 (객실꺼 따로) - 객실과 STAY 다름
OUT	6번 맨좌측 Crew/Staff Sign - 출국 GD+티켓+짐부침 - IMM IN/OUT GD 두장 제출(객실 따로)



Meter/Feet Conversion Table

Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	**43,000 ft**	**12,500 m**	**41,100 ft**
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	**38,100 ft**	**11,300 m**	**37,100 ft**
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	**34,100 ft**	**10,100 m**	**33,100 ft**
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	**30,100 ft**	**8,900 m**	**29,100 ft**
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	**25,600 ft**	**7,500 m**	**24,600 ft**
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	**21,700 ft**	**6,300 m**	**20,700 ft**
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	**17,700 ft**	**5,100 m**	**16,700 ft**
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	**13,800 ft**	**3,900 m**	**12,800 ft**
3,600 m	**11,800 ft**	3,300 m	10,800 ft
3,000 m	**9,800 ft**	**2,700 m**	**8,900 ft**
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	**5,900 ft**	**1,500 m**	**4,900 ft**
1,200 m	3,900 ft		

ALT/HEIGHT Conversion

1,000 m	3,300 ft	500 m	1,600 ft
900 m	**3,000 ft**	**450 m**	**1,500 ft**
800 m	2,600 ft	400 m	1,300 ft
700 m	**2,300 ft**	**350 m**	**1,100 ft**
600 m	2,000 ft	300 m	1,000 ft
550 m	**1,800 ft**		

Difference (ATS A333)

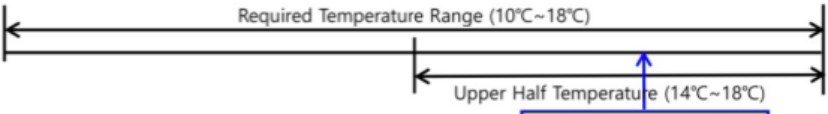
333Q(3)		HL7753, 7754, 7701 C24+Y252(151)	
333R(8)		HL7584-7720 C24+Y260(156)	PACK LO Y Class 60%이하
333R(5) (OZ)	APMS	HL7740-7754 C30+Y260(156)	PACK NOR All Class 200이상
333R(9) (OZ)		HL7792-8293 C30+Y268(160)	
333E(6)	APMS -1.5	HL8001-8027 C24+Y248 **HL8001 RNP AR X**	**RNP AR, OANS, PAX AUTO, TRIM AUTO**

ENG START : Basic 43-48% Enhanced 52-54%
ENG OIL : 12Qt + 0.5Qt/h

Categories

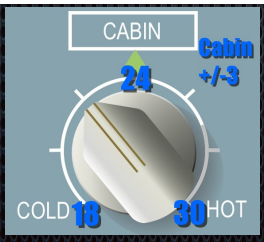
WING SPAN	60.30m		197.83ft
180 Back (72도)	48m		157ft
WING SPAN CAT	ICAO E		FAA V
Wt CAT	Heavy		
APP CAT	ST-IN : C		Circling : C/D(FAA)
524700lbs	**238T**	507063lbs	**230T**
478402lbs	**217T**	436515lbs	**198T**
423287lbs	**192T**		

LOADSHEET : CARGO No. FWD 11-21, AFT 31-41

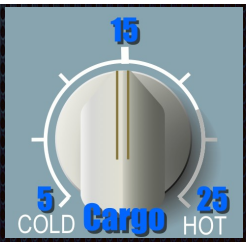


15도 이하시 : TOD -10min 15도

Middle of upper half temperature (16℃)







CRZ FUEL Penalty (Approximation)

A330

4000ft below OPT ALT : 5% increase trip fuel
8000ft below OPT ALT : 10% increase trip fuel
Above OPT ALT : 3% increase trip fuel

FUEL Consumption

APU	GND : 438 LBS/HR (BLOCK FUEL산정) IN FLT : 286 LBS/HR
TAXI	2 ENG, no APU : 3300 LBS/HR (10분 550 LBS)
CRZ	100FT 당 8 MIN ex) 1000FT 상승 80 MIN, 1HR 20MIN
Holding	분당 200LBS (Actual F/F 참조, 시간당 12000LBS)
GA & Landing	3000 LBS (과거 EDTO자료 1회 Missed App+L/D)

FUEL Loading

Tail Tank :
CL TANK 80000LBS 이상 5000LBS 까지
CL TANK 160000LBS 이상 10000 LBS 까지

FUEL Overfill : 1000LBS 기준



ONE ENG TAXI AT ARRIVAL

Taxi with engine 1 is recommended in order to pressurize the blue hydraulic system to ensure brake accumulator pressure

Note: If taxi is performed with engine 2, the flight crew must check that the brake accumulator pressure on the BRAKES and ACCU PRESS indicator is normal.

AT ARRIVAL

APU..... START / CHECK AVAIL F/O

• After high thrust operations:

ENG MIN COOL TIME ...CONSIDER CAPT

Note: The flight crew should operate the engines at or near idle thrust for a cooling period of 5 min before engine shutdown, in order to thermally stabilize the engines.

• When taxiing in a straight line:

ENG 2 SHUTDOWN .....ORDER CAPT

Note: During engine shutdown, a slight jerk forward may occur if the brakes are applied while the aircraft is moving.

ENG 2 ANTI ICE pb-sw ..... OFF F/O

ENG 2 MASTER lever..... OFF F/O

ENG 2 PARAMETERS .....CHECK F/O

X-BLEED ..... OPEN F/O

(To supply both packs from ENG 1)

• **At parking:**

ENG 1 ..... SHUT DOWN CAPT



PACKS USE WITH LP AIR CONDITION UNIT (SD 연결 표시없음)

공항 요구로 APU OFF후 기내 온도 조절을 위한 방법
Low Press Unit을 연결하며 APU와 동시사용 금지

The flight crew must not use conditioned air from the packs and from the LP Air Conditioning Unit at the same time, to prevent any adverse effect on the Air Conditioning system.

절차 POM, FCOM에 없음
APU BLEED OFF 후 LP AIR 연결
EXT PWR 연결후 APU OFF **(AVAIL -> ON)**
APU Start - LP Disconnect - APU BLEED ON

APU BLEED USE WITH HP AIR START UNIT (SD 연결 표시없음)

APU 부작동시 AIR START UNIT으로 시동을 위해 사용
외부 BLEED AIR의 역할을 함. APU BLEED 동시사용 금지

The flight crew must not use bleed air from the APU BLEED and from the HP Air Start Unit at the same time, to prevent any adverse effect on the Bleed Air

- **Before connecting the air start unit:**

PACK 1 & 2 ..... as REQ F/O
HP AIR 시동전 **Air Condition**으로 사용시 ON **(**시동시 **OFF)**
APU BLEED ..... OFF F/O
ENG 1 & 2 BLEED ..... OFF F/O
X BLEED ..... OPEN F/O
AIR START UNIT CONNECT ... REQ CAP
Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.



ENG START
Next Page

STARTING with AIR START UNIT

통상 ENG 1 먼저 **(EXT PWR Start 2 ENG)**
"Req Engine Start up Present Positon~~~"

- **Before connecting the air start unit:**

PACK 1 & 2 ..... OFF F/O
prevent any possible contamination of the packs by the air start unit.
OFF L/T 들어와야함.
APU BLEED ..... OFF F/O
ENG 1 & 2 BLEED ..... OFF F/O
X BLEED ..... OPEN F/O
AIR START UNIT CONNECT ... REQ CAP
Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

- **When cleared to start:**

Note **AIR ABNORM BLEED CONFIG** alert is triggered after the first engine start. It can be disregarded.
ENG 1 ..... START CAP
Note: As necessary, engine 2 can be started first.
In this case, check the brake ACCU pressure prior to engine start.
Note: The minimum recommended starter air supply pressure is 25 PSI when the start valve is open.

- **After engine 1 is started:** AVOD 다 꺼짐
  - **If the air start unit is used to start engine 2:**

ENG 2 ..... START CAP

- **When engine 2 is started:**

AIR START UNIT DISC . . . . REQ CAP
X BLEED ..... AUTO F/O
ENG 1 & 2 BLEED ..... ON F/O
PACK 1 & 2 ..... ON F/O

- **If the crossbleed engine start procedure is used to start engine 2:** EXT PWR – AVAIL first!!

AIR START UNIT(EXT PWR)DISC.RQ ... CAP
ENG 1 BLEED ..... ON F/O
PACK 1 & 2 ..... ON F/O
CRSBLD ENG START PROC.APPLY

EXT PWR ..... CHK AVAIL CAP
EXT PWR DISCONNECT ..... REQ CAP
NOTE : EXT PWR removed after 1 ENG start.
but FCOM : In order to ensure that the entire network remains electrically supplied, it is recommended to start both engines before the disconnection of the external electrical power.
NOR PROC - AFTER START .....RESUME



CrossBleed Start
next page

ENG CROSSBLEED START

#1 ENG BLEED 로 #2 ENG START

Req PushBack

PushBack 완료(Parking Brake set)

One engine must be running in order to supply air for other engine start.

- Before second engine start :

APU BLEED ..... OFF F/O

The BLEED valve of the supplying engine reopens and the cross bleed valve closes.

ENG BLEED (supply engine) ..... ON F/O

ENG BLEED (receive engine) ..... OFF F/O

The bleed valves of receiving engines are closed to avoid reverse flow leakage.

X BLEED..... OPEN F/O

- When cleared to start :

AREA CLEAR OF OBS ..... CONF CAP

Ensure increased power jet wake does not constitute any hazard to people or installations behind the aircraft.

THR LVR (sup eng) . ADJ BLD PRES CAP

Adjust thrust of supplying engine to obtain an engine bleed pressure of **30 PSI**. If the thrust required to obtain the appropriate engine bleed pressure exceeds **40 % N1**, pay particular attention to the surrounding area.

RECEIVING ENGINE ..... START ALL

Apply the normal engine start procedure.

- After start :

THR LVR (supply engine) ..... IDLE CAP

X BLEED..... AUTO F/O

ENG BLEED (receive engine) ..... ON F/O

PACK 1 &2 ..... ON F/O



COLD TEMP CORRECTION General

5도 간격은 보수적으로 보간법 적용됨

Min 제외한 모든 고도 수정은 ATC 인가 필요

Mandatory, Missed App 고도 ATC 사전 인가 없이 금지

반드시 고도 – FE 후의 고도를 보정해야함.

Ex) FE 200ft 공항 : 5000ft는 4800ft만 보정해야함.

Height Above FE (Feet) 200-800ft

TEMP	200	300	400	500	600	700	800
0	20	20	30	30	40	40	50
-5	20	30	40	40	50	60	70
-10	20	30	40	50	60	70	80
-15	30	40	50	60	80	90	100
-20	30	50	60	70	90	100	120

Height Above FE (Feet) 900-5000ft

TEMP	900	1000	1500	2000	3000	4000	5000
0	50	60	90	120	170	230	280
-5	70	80	120	160	230	310	390
-10	90	100	150	200	290	390	490
-15	110	120	180	240	360	480	600
-20	130	140	210	280	420	570	710

GMP, CJU, PUS next page



COLD TEMP CORRECTION (Domestic)

Min 은 반드시 수정 (중간 고도 CORRECTION은 PIC 결정)
Missed App 고도는 ATC 협조 필요

GMP 32L (261') / 32R (262') / 14R (254')

32L/R	8000	5500	5300	4000	2800	2300	2000
0	8450	5810	5600	4230	2970	2440	2120
-5	8620	5930	5710	4310	3030	2490	2160
-10	8780	6040	5820	4390	3080	2530	2200

R14	4000	2800	1400	4000
0	4230	2970	1490	4230
-5	4310	3030	1520	4310
-10	4390	3080	1540	4390

CJU 07 (307') / 25 (296')

	4000	2900	1800	07	8000	25	6000
0	4220	3070	1900		8450		6340
-5	4300	3130	1940		8620		6460
-10	4380	3180	1970		8780		6590

PUS 36L(233'),36R(228') / 18L/R (see below)

36L/R	6000	5000	3300	2100	6000
0	6340	5290	3490	2210	6340
-5	6460	5390	3560	2250	6460
-10	6580	5490	3620	2290	6580

18L/R	6000	5000	4000	2600	1700	6000
0	6340	5290	4230	2760	1800	6340
-5	6460	5390	4310	2810	1830	6460
-10	6580	5490	4390	2860	1870	6580



COLD Wx Operation 1/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :
• visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
• ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PRELIMINARY (OAT 3°C 이하)

PROBE/WIN HEAT – ON (ENG START후 AUTO)

LOGBOOK ---------- CHECK

MAX TIME TO NEXT ENG ACC ---- DETERMINE

TAXI IN TIME – NEXT ENG ACC TIME 15분 이내인지 확인

TAXI-OUT(OAT 3°C 이하)

TAXI TIME ---------- MONITOR

15분 초과 또는 Eng Vibrations시 절차 수행

SOP SURFACE COND & AREA ---------- CHECK

ATC ---------- NOTIFY

PARKING BRAKE ---------- ON

the aircraft starts to move, immediately retard the thrust levers to IDLE.

THR LEVERS ---------- 50% N1 (NO HOLD TIME)

THR LEVERS ---------- RETARD TO IDLE

Repeat this procedure at intervals not longer than 15 min, or if the engine vibrations increase.

SOP TAXI ---------- RESUME

BEFORE TAKEOFF

FLAPS lever ---------- SET FOR TAKEOFF

FLAPS ---------- CHECK PSN

T/O CONFIG pb ---------- TEST

T/O MEMO ---------- CHECK NO BLUE

AFTER START checklist ---------- COMPLETE

BEFORE TAKEOFF checklist --- ACCOMPLISH

TAKEOFF(OAT 3°C 이하)

A final engine acceleration must be performed before takeoff to entirely clean the engine from ice accretion.

ATC ---------- NOTIFY

PARKING BRK(OR BRK PEDALS) ----- APPLY

THR LEVERS ---------- 50% N1 (NO HOLD TIME)

THR LEVERS ---------- RETARD TO IDLE

SOP TAKEOFF ---------- RESUME

AFTER L/D

DO NOT RETRACT FLAPS

Ice Shedding 절차 반복



COLD Wx Operation 2/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :
- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PARKING

After engine shutdown and before shutting down electrical supply:
FLAPS / SLATS ..... CONF FREE OF CONTAM

Perform a visual inspection to determine if the flaps/slats mechanism is free of contamination.
If necessary, perform decontamination.
Y ELEC PUMP pb ..... AUTO F/O
G ELEC PUMP pb ..... AUTO F/O
Y ELEC PUMP pb (guarded) ..... ON F/O
G ELEC PUMP pb (guarded) ..... ON F/O

SLATS / FLAPS ..... RETRACT F/O
Monitor slats/flaps retraction on ECAM upper display
When slats and flaps are retracted:

Y ELEC PUMP pb (guarded) ..... AUTO F/O
G ELEC PUMP pb (guarded) ..... AUTO F/O

NORMAL PROCEDURE ..... RESUME ALL



ENG ON Deicing in ICN

TOBT- 40min CTC KE ICN (사전신청, 결과확인)

ICN Deicing ”Deicing Required ENG On Deicing”
ICN Apron “Req Pushback Deicing Zone xx **SQ2000**“
Pad Control Arrange Deicing Pad No. **REQ FULL BODY**
Ice Man Manage Deicing Process

Gate 시동후 **(FLAP UP, FLT CTL CHK 없이)**
PARKING BRAKE ---------- SET
Report Parking Brake SET - > Ice Man
BROADBAND XMT sw ---------- OFF
COMMUNICATION with GND ------ ESTAB
DE/ANTI-ICING FLUID TYPE ------ CHECK
CABIN PRESS MODE SEL ---- CHK AUTO
ENG 1 & 2 BLEED ---------- OFF
Note : **AIR ENG 1+2 BLEED FAULT** – Disregard
APU BLEED ---------- OFF
DITCHING pb ---------- ON
Note : **CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT OPEN, COND LAV + GAL VENT FALUT** –Disregard
THRUST LEVERS ---------- CHK IDLE
“A/C PREPARED FOR SPRAYING”
Report Ready for Deicing - > Ice Man

Cold Wx

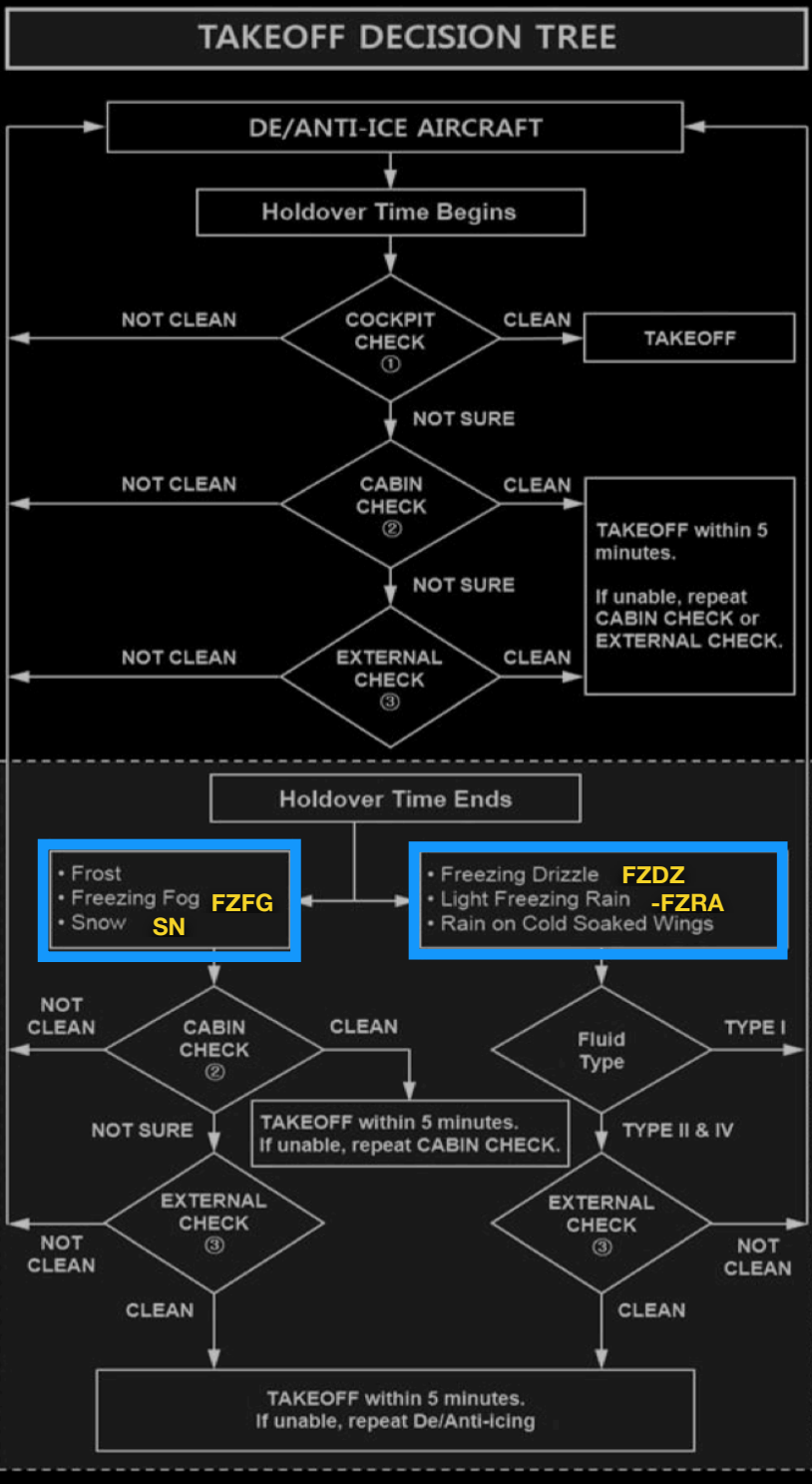
START SPRAY REQ DCL(CTC DEL)

UPON COMPLETE SPRAY(TIME CHK 1분)
용액과 마지막 용액 뿌린 시간 받고 적는다.
Holdover Time 결정!!!
PITOT STATICS (GND Crew) ---------- CHK
GND EQUIPMENT ---------- REMOVE
DITCHING pb ---------- OFF
OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후
ENG 1 & 2 BLEED ---------- ON
BROADBAND XMT sw ---------- ON
5분후
APU BLEED ---------- AS RQRD
NORMAL PROCEDURE ---------- RESUME
FLT CONTROL CHK, FLAPS UP



Decision Tree next page



- External Check = Second Deicing (정비사가 보고 판단해야함)

- Heavy Ice Pellets(**+PL**), Heavy Snow(**+SN**), Hail(Small Hail 제외)(**GR**), Moderate 또는 Heavy Freezing Rain(**FZRA, +FZRA**) 이나 Heavy Freezing Drizzle(**+FZDZ**) 이 공항에 내리는 경우 이륙을 금지한다.

ENG OFF Deicing in GMP...

TOBT- 20min CTC KE GMP (PAD, New TOBT)

REQ DCL
Deicing "Deicing Required PADxxx" 5min TOBT
Apron "Req Pushback Deicing PADxxx"

Gate 시동후 (FLAP UP, FLT CTL CHK, APU OFF 없이)
Engine anti-ice sw ---------- OFF
PARKING BRAKE ---------- SET
BROADBAND XMT sw ---------- OFF
COMMUNICATION with GND ------ ESTAB
DE/ANTI-ICING FLUID TYPE ------ CHECK
CABIN PRESS MODE SEL ---- CHK AUTO
(HL7524-7720) GND COOL ---------- OFF
ENG 1 & 2 BLEED ---------- OFF
APU BLEED ---------- OFF
DITCHING pb ---------- ON
Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT OPEN, COND LAV + GAL VENT FALUT –Disregard
ENG MASTER sw ---------- OFF
SHUTDOWN CHECKLIST
"A/C PREPARED FOR SPRAYING"

UPON COMPLETE SPRAY(TIME CHK 1분)
용액과 마지막 용액 뿌린 시간 받고 적는다.
Holdover Time 결정!!!

Cold Wx

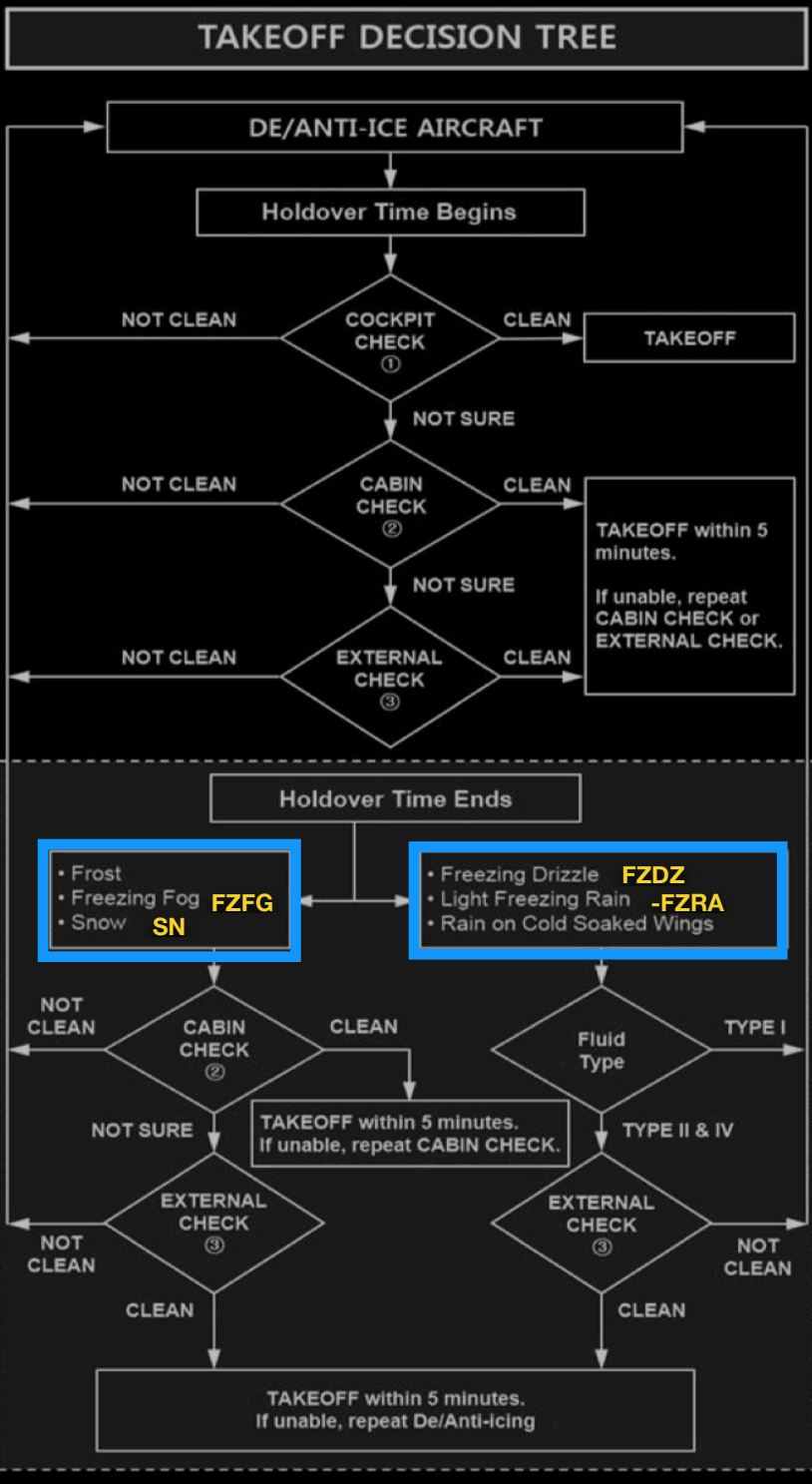
(HL7524-7720) GND COOL ---------- OFF
PITOT STATICS (GND Crew) --------- CHK
GND EQUIPMENT ---------- REMOVE
DITCHING pb ---------- OFF
OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후
ENG 1 & 2 BLEED ---------- ON
BROADBAND XMT sw ---------- ON
5분후
APU BLEED ---------- ON
NORMAL PROCEDURE ---------- RESUME
Note : INIT B check & re-enter
PREFLT CHKlist -> Req STARTUP -> CHKlist

BOTH ENGINES ARE STARTED
FLT CTL CHK, FLAPS UP, APU as REQ



Decision Tree next page



- External Check = Second Deicing (정비사가 보고 판단해야함)

- Heavy Ice Pellets(**+PL**), Heavy Snow(**+SN**), Hail(Small Hail 제외)(**GR**), Moderate 또는 Heavy Freezing Rain(**FZRA, +FZRA**) 이나 Heavy Freezing Drizzle(**+FZDZ**) 이 공항에 내리는 경우 이륙을 금지한다.

QRH LIST Reference Only

AIR	BLEED 1+2 FAULT(APU BLEED ON) BLEED 1+2 FAULT(APU BLEED NOT AVAIL)
DOOR	COCKPIT DOOR FAULT
EIS	DISPLAY UNIT FAILURE EIS DISPLAY DISCREPANCY
ELEC	C/B TRIPPED **ELEC EMER CONFIG SYS-REMAINING**
ENG	**ALL ENG FAIL** ENGINE FUEL SYS CONTAMINATION ENG RELIGHT IN FLIGHT ENG STALL ENGINE TAILPIPE FIRE HIGH ENGINE VIBRATION ON GND - NON ENG SHUTDN AFT ENG OFF ONE ENGINE INOPER - CIRCLING APP ENG1(2) EGT OVERLIMIT
F/CTL	LANDING WITH SLATS OR FLAPS JAMMED NO FLAPS NO SLATS LANDING RUDDER JAM / RUDDER PEDAL JAM RUDDER TRIM RUNAWAY
FUEL	FUEL IMBALANCE FUEL LEAK FUEL LOSS REDUCTION FUEL OVERREAD GRAVITY FUEL FEEDING TRIM TANK FUEL UNUSABLE
L/G	LANDING WITH ABNORMAL L/G L/G GRAVITY EXTENSION
NAV	ABNORMAL V ALPHA PROT ADR 1+2+3 FAULT UNREL SPD INDICATION IR ALIGNMENT IN ATT MODE NAV FM / GPS POS DISAGREE **ALL ADR OFF**
SMOKE	**SMOKE / FUMES / AVNCS SMOKE** **REMOVAL OF SMOKE / FUMES** SMOKE / FIRE FROM LITHIUM BATTERY
WHEEL	WHEEL TIRE DAMAGE SUSPECTED
MISC	BOMB ON BOARD COCKPIT WINDSHIELD/WINDOW ARCING COCKPIT WINDSHIELD/WINDOW CRACKED DITCHING EMER EVAC EMER LANDING FORCED LANDING OVERWEIGHT LANDING SEVERE TURBULENCE TAILSTRIKE VOLCANIC ASH ENCOUNTER



SYSTEM RESET TABLE 1/2

CKPT CTL pb 3초 **RESET BTN 1초**

A-ICE	**2 WHC**	GND : A.ICE L(R) WSHLD(WINDOW) HEAT
AIR	**ENG BLEED pb**	AIR ENG 1(2) BLEED FAULT GND. : AIR ENG 1(2) BLEED NOT CLSD
	2 PACK CONT	PACK CONT RESET
COND	**1 ZONE CONT**	ZONE CONT RESET
APU	**APU MASTER**	Only APU SPD below 7%
AUTO FLT	**REF FCOM**	GND/Before TAXI : AUTO FLT A/THR OFF
	2 FCU	PART FCU, FCU
	2 FMGEC 2 FM	AUTO FLT FM 1(2) FAULT MAP NOT AVAIL
	MCDU OFF	MCDU RESET
BRKS	**REF FCOM**	GND : BRAKES RESIDUAL BRAKING WHEEL N/W STRG FAULT
CAB PR	**2 CPC**	FAULTY CPC RESET
COM	**2 CIDS**	Cabin Intercom Data Sys RESET
	2 ACP/AMU	ACP RESET
	RMP OFF	RMP RESET
	ELT RESET	ELT RESET
DATA LINK	**1 ATSU**	INVALID DATA or ATSU RESET
	1 AINS,1 CINS	AINS, CINS RESET
DOOR	**2 PSCU**	Proxi Switch Cont Unit RESET
EIS	**DU OFF**	ECAM, PFD, ND RESET
	3 DMC	DMC RESET
ELEC	**EXT A,B OFF**	GND : RESET EXT A,B
	3 BAT pb OFF	BATT RESET
	COMM, GALL pb OFF	GND : COMMER, GALL RESET
ENG	**2 EIVMU**	Eng Interface Vibration Monitor Unit RESET



SYSTEM RESET TABLE 1/2

CKPT CTL pb 3초 **RESET BTN 1초**

	PRIM,SEC OFF	GND : PRIM, SEC RESET
F/CTL	**2 FCDC**	FCDC RESET
	2 FLAP	GND : F/CTL FLAP SYS FAULT
	2 SLAT	GND : F/CTL SLAT SYS FAULT
FUEL	**2 FCMC**	GND : FUEL ZFW ZFCG DISAGREE
FWS	**2 FWC, 2 SDAC**	Waring Sys RESET
L/G	**2 LGCIU**	LGCIU RESET
MISC	**VSC (Cabin aft C/B)**	RESET VACUUM TOILET SYS (Blue pb)
COND	**1 ZONE CONT**	ZONE CONT RESET
SMOKE	**2 SDCU**	Smoke Dectect RESET
VENT	**2 VENT CONT**	RESET VENT CONT
	1 AEVC	RESET VENT COMPUTER

